



State Estimation for Integrated Energy Systems: Motivations, Advances, and Challenges

PREPARED BY THE

IEEE PES Energy Internet Coordinating Committee

IEEE PES Working Group on Power System Static and Dynamic State Estimation

Task Force on State Estimation for Integrated Energy Systems

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KEYWORDS

Control Centers, Dynamic State Estimation, Distribution Systems, Distributed Energy Resources, Energy Internet, Energy Management Systems, Measurement, Integrated Energy System, Robust Estimation, Renewable Energy, Machine Learning, Bad Data, State Estimation.

ACRONYMS

AMI	Advanced Metering Infrastructure
ADMM	Alternating Direction Method of Multipliers
CCHP	Combined Cooling Heating and Power
CHP	Combined Heat and Power
CWLS	Constrained Weighted Least Square
DSE	Dynamic State Estimation
DHS	District Heating System
ECM	Electric Circuit Method
EPS	Electric Power System
EKF	Extended Kalman Filter
EMS	Energy Management System
FDDP	Frequency Domain Distributed Parameter
HSOC	Heat State of Charge
ICCP	Inter-Control Center Communications Protocol
IEHS	Integrated Electricity-heat System
IESs	Integrated Energy Systems
IEMS	Integrated Energy Management System
IES-DSE	Integrated Energy Systems - Dynamic State Estimation
IES-SE	Integrated Energy Systems - State Estimation
IES-SSE	Integrated Energy Systems - Static State Estimation
ISOs	Independent System Operators
KF	Kalman Filter
NGS	Natural Gas System
PDE	Partial Differential Equation
PMU	Phasor Measurement Unit
PS-SE	Power System State Estimation
SCADA	Supervisory Control and Data Acquisition
SE	State Estimation
SSE	Static State Estimation
STP	Standard Temperature and Pressure
TDPP	Time Domain Distributed Parameter
TF	Task Force
WLAV	Weighted Least Absolute Value
WLS	Weighted Least Square

TABLE OF CONTENTS

1. Task Force Scope	1
2. Background Information.....	1
3. Introduction to IESs.....	1
3.1 Motivations for IESs	1
3.2 Definitions of IESs	3
3.3 Characteristics of IESs.....	4
3.4 Modeling of IESs	6
3.4.1 Electric Power Systems Modeling	6
3.4.2 Heat System Modeling.....	7
3.4.3 Natural Gas Modeling.....	13
3.4.4 Energy Circuit Method for Heterogeneous Energy System	16
3.4.5 Modeling Differences between Integrated Energy Systems - State Estimation and Traditional Tools	21
3.4.6 Coupling Device Modeling	22
3.5 Necessity of Integrated Energy Systems-Energy Management System	23
4. State Estimation of IESs.....	24
4.1 Motivations for Integrated Energy Systems-State Estimation	24
4.2 Similarities and Differences between Integrated Energy Systems-State Estimation and Power System-State Estimation	24
4.3 Integrated Energy Systems-State Estimation Models (Multi-time Scales)	27
4.3.1 Actual Value vs Per-unit Value.....	27
4.3.2 State Variables in State Estimation of IESs	27
4.3.3 State Estimation Models for IESs: High-level Framework	27
4.4 Integrated Energy Systems-State Estimation Measurement Placement and Types	30
4.4.1 Measurement Types.....	30
4.4.2 Measurement Placement.....	31
4.5 Integrated Energy Systems-State Estimation Observability Analysis.....	31
4.6 Integrated Energy Systems-State Estimation Methods	34
4.6.1 Based on Problem Formulation	34
4.6.2 Based on Computing Architecture	36
4.6.3 Consideration of Uncertainties	37

4.6.4	Consideration of Different Sampling Resolutions	37
4.6.5	Data Exchange Between Individual Systems	38
4.7	Difference between Integrated Energy Systems-State Estimation Algorithms	39
4.7.1	Modeling Differences	39
4.7.2	Measurement Differences	40
5.	Demonstration Project in China	40
5.1	System Description	41
5.2	Test Results in an Industrial Park	42
5.3	Test Results in a City-Scale IES	45
6.	Future Works and Open Challenges	46
7.	References	47

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1. Task Force Scope

This Task Force (TF) report aims to:

- Identify the theoretical barriers of State Estimation (SE) for Integrated Energy Systems (IESs), such as modeling, observability analysis/measurement requirements, addressing multiscale dynamics and different sources of energy, wide engagements of ubiquitous communication resources, as well as their practical applications.
- Gather industrial perspectives and provide solutions and recommendations to vendors, national labs, utilities, and Independent System Operators (ISOs) in using SE for IESs modeling, monitoring, operation, and control.
- Educate the next generation of power engineers for Integrated Energy Systems – State Estimation (IES-SE) and applications and students in learning and using new tools.

2. Background Information

IESs consisting of electricity, heating, cooling, natural gas, and transportation systems, form the foundational physical layer in the energy internet. The modeling and awareness of their real-time operating states are therefore indispensable for energy internet operation. IESs have existed for many years, e.g., Combined Heat and Power (CHP) productions account for 30-50% of the total power generation in the winter in Northern Europe and Northern China. However, they must be effectively coordinated, which requires extended situational awareness. In bulk power systems, SE delivers situational awareness and thus has been a key foundational component of today's energy management system (EMS). Although there have been studies on SE models and algorithms for heating and natural gas systems, SE for general IESs is still challenging due to:

- Higher complexity than the bulk power systems with multiple energy sectors models and time scales.
- Distributed systems with multiple system operators and multi-energy consumers.

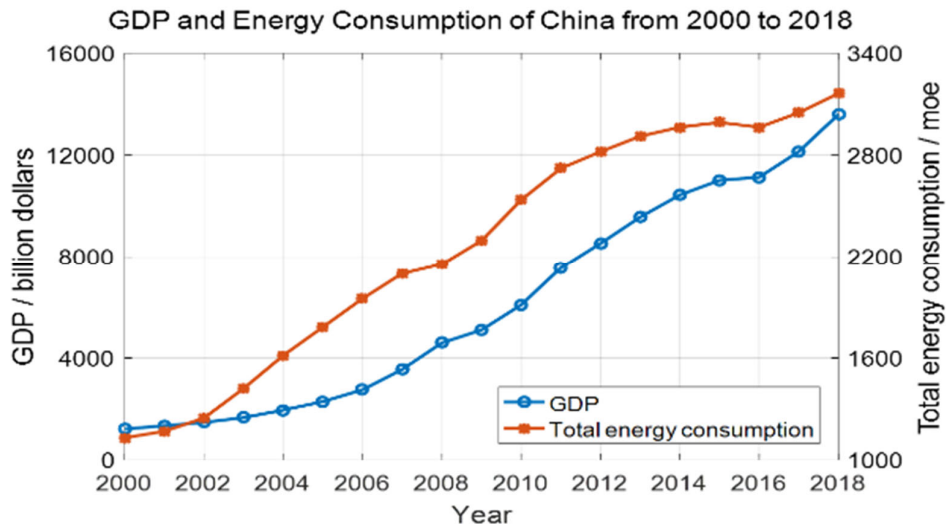
The existing theories and experiences of SE in bulk power systems or other engineered systems are not directly applicable to IESs. Therefore, it is of crucial importance to form a new TF which addresses both theoretical and practical barriers for IESs and establishes a solid cornerstone for further developments of energy internet operation, optimization, and control.

3. Introduction to IESs

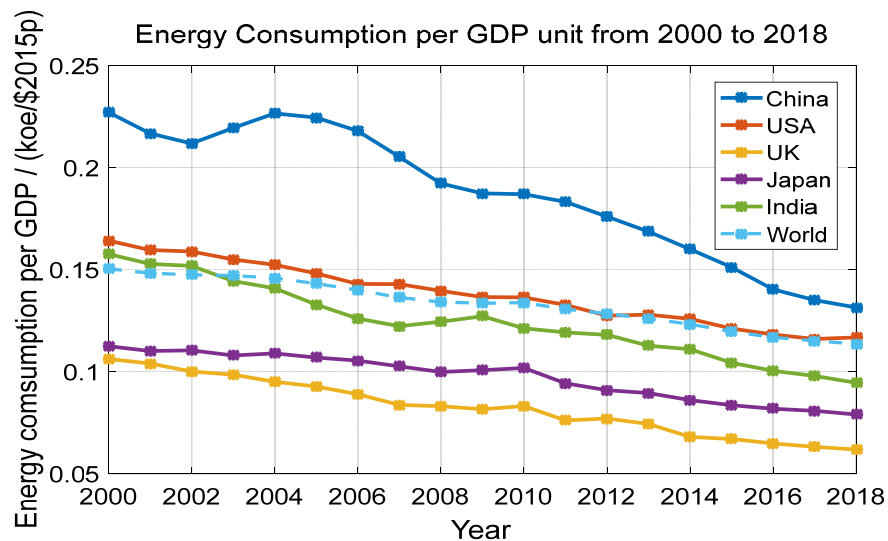
3.1 Motivations for IESs

The energy crisis is one of the most significant challenges faced in the 21st century. On one hand, the energy demand is rapidly growing, not only in quantity but also in terms of quality, convenience, and environmental friendliness. Considering China as an example, as shown in *Figure 1 (a)*, together with its rapid economic development, its energy consumption has almost tripled in the last decade. On the other hand, traditional fossil fuel resources are being depleted, while the integration of renewable energies brings challenges of uncertainties and volatility. *Figure 1 (b)* shows that energy efficiencies in many countries have stagnated in recent years. These problems indicate that traditional roadmaps of research and development are gradually

reaching their bottlenecks and more fundamental changes are needed for a sustainable future.



(a) Growth of GDP and energy consumption in China [1]



(b) Energy efficiency in some countries and the world average [2, 3]

Figure 1: Rapid increase in energy consumption and stagnating efficiency of the energy system.

Energy is essential for sustaining modern societies. Nowadays, energy, environment, and sustainable development are the hottest spots of common concern in the world [4, 5]. With the development of the economy and society, energy demand is constantly increasing. However, environmental pollution in the process of energy use affects and restricts the development of the economy and society. Traditional fossil energy's non-renewable nature and ecological pollution poses limits to their future uses. In order to achieve the sustainable development of the economy and society, it

is an inevitable choice for human society to improve energy efficiency, explore new energy sources, and achieve energy conservation and emission reduction [6, 7].

Apart from electricity, the primary forms of energy used in human society include gas and heating/cooling. In the traditional energy system, all kinds of energy systems are planned, designed, and operated independently, despite their close physical couplings. For example, in northern Europe and China, CHP generation accounts for 30-50% of the total electricity production in the winter [1, 2]. Another example is the increasing interdependence between the electric and natural gas sectors [3]. It is driven by the low cost of natural gas-fired generating units. Moreover, many natural gas-fired units can provide operating flexibility which is required with the high penetration of renewable energy sources. The independent operation of these energy sectors leads to low efficiency of energy use. In order to improve the efficiency of energy use and the hosting capacity of renewable energy, there is an increasingly urgent need for interconnection and integration of various energy systems. At the same time, power systems, gas systems, heating systems, and transportation systems have the complementarity of physical characteristics (for example, electric energy is easy to transmit but difficult to store, thermal energy is difficult to transmit but easy to store; and natural gas is easy to transmit and store). There are obvious staggered peak-valley patterns among various energy loads such as electricity, gas, and heating/cooling. This phenomenon makes the combined system of electricity, gas, and heating/cooling have more control means, more adjustment space, and better economy of the combined operation. Moreover, various kinds of energy conversion equipment in industrial production (such as CHP, Combined Cooling Heating and Power (CCHP), electric heating, heat pump, fuel-cells, electric vehicle charging, and/or electric hydrogen generation) also provide means for the interconnection of various energy sources. Hence, an important direction of breaking through is the coordination of different energy sectors, such as electricity, heat, and natural gas.

Note that the strong complementary properties of the different energy sectors can be leveraged for more efficient and effective operation. For example, with proper energy conversion devices, electricity can serve as a fast-heating supply and the heating system can in turn help to offset electric power fluctuations with its high inertia. Similarly, in electric power systems, there is usually low load demand and high wind power at night, sometimes leading to wind curtailment. At the same time, the heating demand may reach its peak. This difference in demand profiles can be used to optimize both the systems. In summary, considering these physical couplings and complementary properties, IESs must be studied from a holistic point of view. Further, these reasons make IESs/multi-energy flow systems/energy internet a development trend in the energy field in recent years.

3.2 Definitions of IESs

The scope of an IES may include but is not limited to, an electric power system, heat system, natural gas system, cooling system, and transportation system. These energy sectors in Energy Internet are illustrated in *Figure 2*. These energy sectors are interdependent and connected via different energy conversion devices e.g., CHP units, gas turbines, electric boilers, electric vehicle charging stations, and/or hydrogen production [8-10].

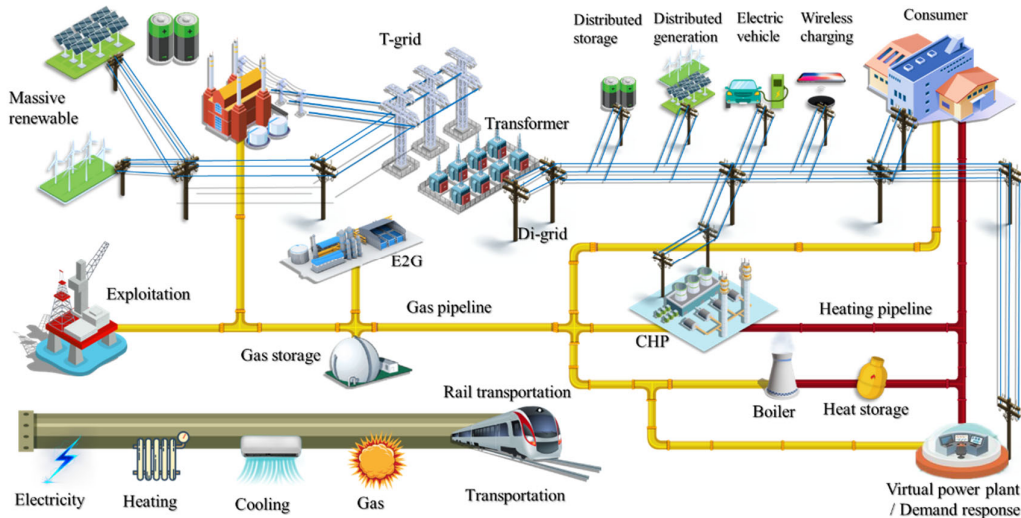


Figure 2: Structure of an IES.

The development of IES, i.e., the coordination of different energy sectors, can achieve the following benefits:

- Exploiting synergies and complementary advantages of various energy sectors for system design and operation.
- Carbon emission reduction by increasing the whole system's energy efficiency and flexibility.
- Facilitating the integration of local sustainable and renewable energy resources.
- Reduction of operational or delay of capital expenditure on energy systems.
- Cost-effective provision of flexibility to the electric power system.
- Improved system reliability and resilience.
- Opportunities for business innovation [10].

3.3 Characteristics of IESs

A typical IES with electricity, heat, and gas is shown in *Figure 3*. The IES is composed of a power system, thermal system, natural gas system, transportation systems, and energy conversion equipment acting as coupling links [11]. The power system is responsible for the production, transmission, distribution, and consumption of electric energy. For the thermal system, the high-temperature water generated by the heat source is transmitted to the heat load through the heating network, also referred to as district (or industrial) heating systems. The high-temperature water after passing through the radiator of the heat load becomes low-temperature water and then flows back to the heat source through the regenerative network. It is expected that in the future heat pumps will have to play a major role in the coupling of electric and heating sectors. Heat pumps can use renewable electricity and are much more efficient than gas-fed boilers. This serves as a motivation for the planned deployment of heat pumps, probably in combination with low-temperature (i.e., underground) geothermal energy at high latitudes in some parts of Europe. In the case of the gas system, it is responsible for the production (from the gas source), delivery (from the supply pipeline), and consumption (from the gas load) of natural gas. To ensure pressure, the natural gas

system also needs a compressor. A highly electrified transportation sector can also play an important role in IES. Rail transportation and electric vehicles will become major load components in electric power systems, and vehicle-to-grid (V2G) technologies allow electric vehicle fleets to act as portable, distributed energy storage systems. The CHP units, CCHP units, electric boilers, fuel cells, gas boilers, and electric vehicle chargers are used to complete the conversion between different energy sources.

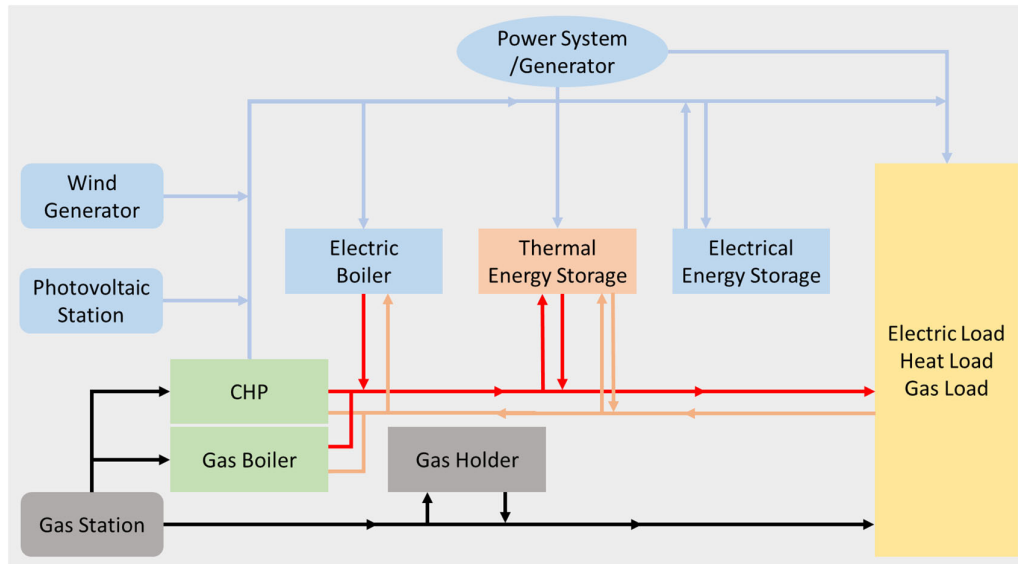


Figure 3: Schematic diagram of an IES with electricity, heat, and gas.

As shown in Figure 3, the main characteristics of IES including the power system, thermal system, and gas system are as follows [6]:

- IES involves different physical quantities in the multi-energy domain (electricity, heating/cooling, and gas), each of which follows different physical laws. For example, the power system follows the laws of electromagnetism and circuit [12, 13]. The thermodynamic system (heating/cooling system) includes hydraulic and thermodynamic models, and follows the laws of fluid mechanics and thermodynamics [14, 15]. The natural gas systems follow the laws of fluid mechanics [16, 17].
- Multi-energy flow coupling has a high degree of physical heterogeneity. The physical characteristics of the power system, thermal system, and natural gas system are essentially different. They belong to multiple heterogeneous energy flow systems, coupled together through CHP, CCHP, electric heating, heat pump, electric hydrogen production, and other equipment.
- Each system in IES has different dynamic processes with varying scales of time as shown in Figure 4. The time constant of the power system is the smallest and the rate of change of system operating conditions is the fastest, the time constant of the thermal system and the natural gas system is in the median scale. The above phenomena lead to the multi-time scale characteristics of IES.
- Different systems in IESs are typically owned and operated by different entities, which leads to barriers for information exchange and operation coordination. In most regions in the world, the electricity, heat, and natural gas that comprise

IES belong to different companies and management bodies. So there are industry barriers such as information privacy, operational differences, and objective conflicts [6], which bring challenges to integrated management and scheduling. These barriers are less significant for centralized IES.

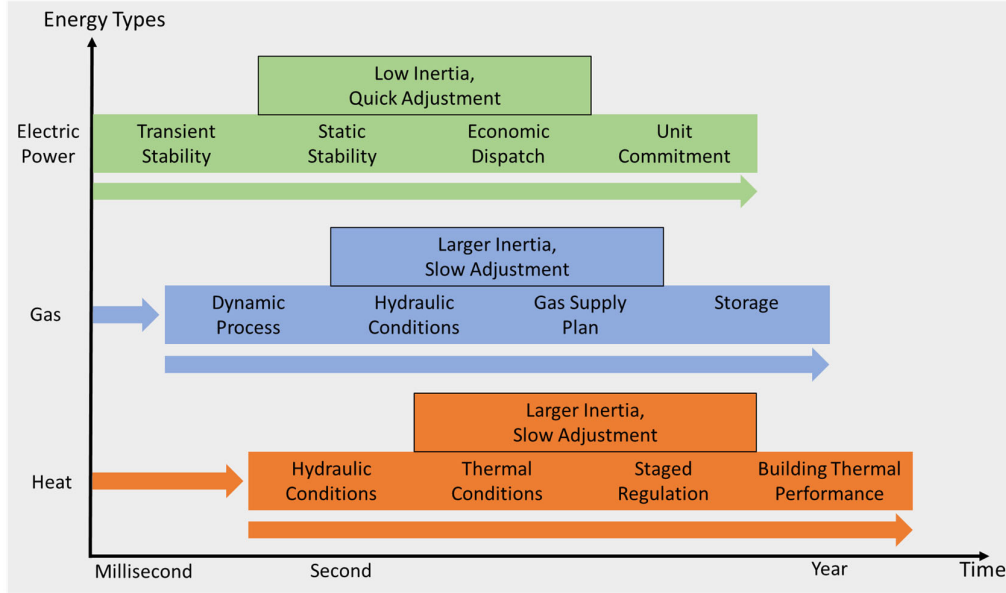


Figure 4: Schematic diagram of multiple time scale characteristics of IES.

3.4 Modeling of IESs

Considering their different properties, different energy sectors are modelled in heterogenous time scales as described below.

3.4.1 Electric Power Systems Modeling

The steady-state operation of electric power systems follows AC power flow equations [18]:

$$\begin{aligned}
 P_i &= \sum_{j=1}^N V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}) \\
 Q_i &= \sum_{j=1}^N V_i V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij})
 \end{aligned} \tag{1}$$

where voltage magnitude V_i and phase angle θ_i are state variables of node i , while P_i and Q_i are its real power and reactive power injections, respectively. Matrices $\mathbf{G}$ and $\mathbf{B}$ are real and imaginary parts of the bus admittance matrix, respectively.

In SE problem, branch power measurements are also used. The relation between branch power flows and nodal voltages is given by:

$$\begin{aligned}
 P_{ij} &= V_i^2 g - V_i V_j g \cos \theta_{ij} - V_i V_j b \sin \theta_{ij} \\
 Q_{ij} &= -V_i^2 (b + y_c) - V_i V_j g \sin \theta_{ij} + V_i V_j b \cos \theta_{ij}
 \end{aligned} \tag{2}$$

where g , b , and y_c are branch serial conductance, serial susceptance, and shunt susceptance, respectively.

3.4.2 Heat System Modeling

This section presents the steady-state hydraulic model and the dynamic thermal model. Transients in the hydraulic system (say pressure variations) are faster than the shortest sampling period of the conventional measurements used in SE of heat networks. This is the reason why modeling dynamics of hydraulic system is inexpedient in SE even if sampling frequency is relatively high. On the other hand, temperature variations are much slower, and they can be captured by widely used measurement system in heat networks. State variables in the heating system usually include node/pipe mass flow rates and node pressures in the hydraulic model, as well as node temperatures in the supply and return networks, node exchanger supply and return temperatures, and pipelines' inlet/outlet temperatures in the thermal model.

a) Hydraulic Model

The hydraulic model describes the relationship between the mass flow rates and pressure distribution in the heating system. The hydraulic model is based on physical laws analogous to Kirchhoff's law [19] [20], which is similar to the relations in the electric power system as shown in *Table 1*.

Table 1: The comparison of Kirchhoff's law in hydraulic systems and electric power systems

Kirchhoff's law in	Node	Loop	Single branch (Circuit or pipe)
Electric power system	Kirchhoff's current law	Kirchhoff's voltage law	Ohm's law
Hydraulic system	Continuity of flow equation	Pressure drop equation	Head loss equation

Let us consider two points (p, x) and $(p + dp, x + dx)$ along a pipe with constant cross-section A , where p is pressure, x is horizontal axis, and z is the vertical axis. Bernoulli's equation can then be expressed in the form [21]:

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = \frac{p + dp}{(\rho + d\rho)g} + \frac{(v + dv)^2}{2g} + (z + dz) + dh_f \quad (3)$$

where ρ is fluid density, g is acceleration due to gravity, v is the fluid velocity, and z is the distance above the horizontal axis. Each term in equation (3) represents energy per unit weight of fluid and has a dimension of length; the first term models work done by pressure, the second is kinetic energy, and the third is potential energy. The sum of these three terms is always constant and is called head. Therefore, the term dh_f gives the head loss due to fluid friction in the pipe section dx . The following assumptions are then introduced [22, 23]:

- Fluid water flow is in a steady state and fully developed. This means there is no water motion in the radial direction, and the axial velocity component is constant.
- The water flow in a pipe is modelled via the lumped mass model. This is a "black box" pipe model, where spatial variation of quantities and parameters along pipes is not considered.

- Given the temperature change in pipes is relatively narrow, the properties of water do not change with temperature in hydraulic model, so that temperature is not involved in the construction of the hydraulic model.
- Variation of water density with pressure and temperature in the operating range is negligible; the water is incompressible.
- For the range of operating temperatures and pressures, dynamic and kinematic viscosities are approximately constant.
- Friction factor $f(\cdot)$ is constant along a pipeline and pipe segments.
- Variation of the supply temperature at each heat source is negligible and heat pipes do not have storage capability.

Water flow physics in heat pipes is similar to the gas flows, as discussed later in Section 3.4.3. In this case, mass flow rates $\dot{m}$ can be used instead of volumetric flow rates q to describe the flow in pipes. The right-hand side of equation (3) describes the head loss (or, loss of pressure) in the heat pipes and the sum of head losses within each loop l equals zero:

$$\sum_{b_l=1}^{B_l} h_{b_l}(\cdot) = 0, \quad l = 1, 2, \dots \quad (4)$$

where index b_l denotes branch/pipe b in loop l ; B_l is the total number of pipes in loop l ; h_{b_l} is the head loss of pipe b in loop l , which is calculated using the Darcy-Weisbach relation [24]:

$$h_{b_l} = K_{b_l} \dot{m}_{b_l} |\dot{m}_{b_l}|; \quad K_{b_l} = \frac{8L_{b_l} f_{b_l}(\cdot)}{D_{b_l}^5 \rho^2 \pi^2 g} \quad (5)$$

where L_{b_l} and D_{b_l} are the length and diameter of pipe b_l , while water friction $f_{b_l}(\cdot)$ is dependent on Reynolds number and, consequently, flow type (e.g., laminar, turbulent) [25]. Also, product $\dot{m}_{b_l} |\dot{m}_{b_l}|$ specifies head loss direction.

If there is no leakage in pipes, the continuity of water mass flow rates at node i is given by:

$$\sum_{b_i^{in}=1}^{B_i^{in}} \dot{m}_{b_i^{in}} = \dot{m}_{load_i} + \sum_{b_i^{out}=1}^{B_i^{out}} \dot{m}_{b_i^{out}}, \quad i = 1, 2, \dots \quad (6)$$

where b_i^{in} and b_i^{out} are indices of branches incoming to and outgoing from node i , respectively, and B_i^{in} and B_i^{out} are the total numbers of branches incoming to and outgoing from node i , respectively. Equation (6) holds for both supply and return nodes; in the latter case, $\dot{m}_{load_i}$ comes with the negative sign.

b) Thermal Model

A heating system consists of heat exchangers, a supply network, and a return network. Each heat exchanger is connected to a pair of locations, one in the supply and the other in the return network. Supply and return networks consist of pipelines as shown in Figure 5; it was assumed that each node has a heat exchanger terminal connected to it. The terms τ, T, T^N are used to denote the pipeline temperatures, locational temperatures, and heat exchanger's temperatures, respectively, in the following

explanation. More specifically, $\tau_{ij}^{S/R}(x, t)$ denotes the temperature of pipeline ij at a distance x from its inlet at time period t in supply/return network (accordingly, $\tau_{ij}^S(L, t)$ and $\tau_{ij}^R(L, t)$ is outlet temperature of the pipeline ij with length L at time t in supply and return network, respectively); $T_{i,t}^S$ and $T_{i,t}^R$ is locational temperature at node i , time t in supply/return network; and $T_{i,t}^{NS}$ and $T_{i,t}^{NR}$ is heat exchanger's temperature at node i , time t in supply/return network.

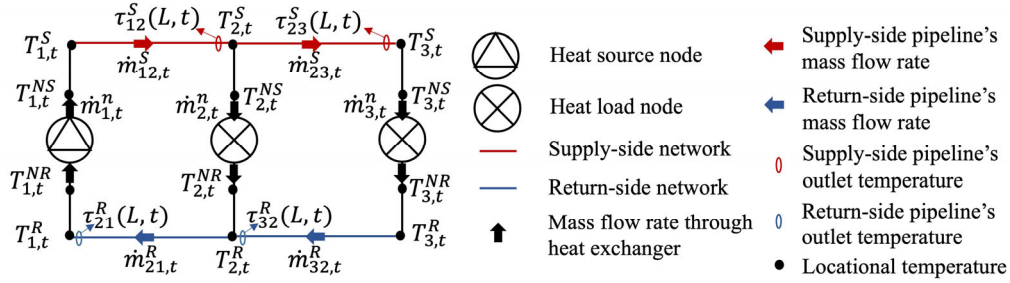


Figure 5: Illustration of a heat system's part including source node 1 and load nodes 2&3.

Load nodes use heat exchangers to get heat from high-temperature water in the supply network and release low-temperature water to the return network. Source nodes reheat low-temperature water in the return network and send out high-temperature water to the supply network.

Heat power ϕ_i consumed/produced at consumer/source node i at period t is obtained as the nodal heat power constraint [10], [11]:

$$\phi_{i,t} = C_p \dot{m}_{i,t}^n (T_{i,t}^{NS} - T_{i,t}^{NR}) \quad (7)$$

where $T_{i,t}^{NS/NR}$ denotes heat exchanger's supply/return-side temperatures of node i at period t , C_p is the specific capacity of water, and $\dot{m}_{i,t}^n = \frac{dm_{i,t}^n}{dt}$ represents the mass flow rate through node i 's heat exchanger at period t .

Since the dynamic time constants of the thermal process in the heating system vary from several minutes to hours, both the static and dynamic thermal models are formulated in this section.

The starting point for both approaches is the initial partial differential equation (PDE). In the static temperature model, temporal transients are not modelled and the spatial coordinate " x " is integrated out. On the other hand, dynamic models contain both temporal and spatial variations. These cases are discussed below.

Static Thermal Model

The following assumptions are considered [26]:

- The heat transfer rate between the fluid in a pipe and the pipe external surface is driven by both convective and conductive transfers; these two heat transfers are modelled with the overall heat transfer coefficient λ , which is constant along a pipe.

- The specific heat capacity of water ($W/kg^\circ K$) is in general dependent on initial pressure p and temperature T , i.e., $C_p(p, T)$. It can be considered constant in heat pipes where the ranges of pressure and temperature changes are not high.

Consider a pipe of length dx , where the rate of heat transfer from heated water to the pipe ambient in radial direction is $\frac{d\phi}{dt} = (T - T^a)\lambda dx$, where T is the temperature of heated water, T^a is the ambient water temperature and λ is the overall heat transfer coefficient of pipe including conductive and convective heat transfers. The heat loss $d\phi$ in a pipe segment is $d\phi = C_p \times dm \times dT$, where C_p is the specific capacity of water, dm is the unit mass of water, and dT is the temperature change. The combination of two relations above yields the differential equation:

$$\frac{dT}{(T - T^a)} = \frac{\lambda dx}{C_p \dot{m}} \quad (8)$$

To eliminate spatial variation, (8) is integrated over pipe x -axis from $x = 0$ to $x = L$. This gives the temperature at the end ($x = L$) of each pipe ij in supply/return networks:

$$\tau_{ij,t}^{S/R}(L, t) = (T_{i,t}^{S/R} - T_{i,t}^a) \times e^{-\frac{\lambda_{ij}^{S/R} L_{ij}^{S/R}}{C_p \dot{m}_{ij,t}^{S/R}}} + T_{i,t}^a \quad (9)$$

where i and j are the inlet and outlet nodes, respectively, of the considered pipe ij , $\lambda_{ij}^{S/R}$ is the overall heat transfer coefficient in the supply/return-side pipe ij , $L_{ij}^{S/R}$ is the length of supply/return-side pipe ij , $\dot{m}_{ij,t}^{S/R} = \frac{dm_{ij,t}^{S/R}}{dt}$ represents the mass flow rate in the supply/return-side pipe ij and subscript t denotes the time interval in which steady-state is studied.

In heat networks, heat balance at each node must be maintained. There are two kinds of heat nodes including heat source nodes and heat load nodes. Assuming that the mixture of heat water at the considered node i is perfect and there is only one output temperature, the temperature mixing processes can be given in terms of mass flow rates. For heat source node $i \in \psi_G$, where ψ_G represents the set of heat source nodes, it follows that:

$$(\dot{m}_{i,t}^n + \sum \dot{m}_{ji,t}^S) T_{i,t}^S = \dot{m}_{i,t}^n T_{i,t}^{NS} + \sum \dot{m}_{ji,t}^S \tau_{ji,t}^S(L, t) \quad (10)$$

$$(\sum \dot{m}_{ji,t}^R) T_{i,t}^R = \sum \dot{m}_{ji,t}^R \tau_{ji,t}^R(L, t) \quad (11)$$

$$T_{i,t}^R = T_{i,t}^{NR} \quad (12)$$

For heat load node $i \in \psi_L$, where ψ_L represents the set of heat load nodes, the following comes out:

$$(\dot{m}_{i,t}^n + \sum \dot{m}_{ji,t}^R) T_{i,t}^R = \dot{m}_{i,t}^n T_{i,t}^{NR} + \sum \dot{m}_{ji,t}^R \tau_{ji,t}^R(L, t) \quad (13)$$

$$(\sum \dot{m}_{ji,t}^S) T_{i,t}^S = \sum \dot{m}_{ji,t}^S \tau_{ji,t}^S(L, t) \quad (14)$$

$$T_{i,t}^S = T_{i,t}^{NS} \quad (15)$$

where $\dot{m}_{i,t}^n = \frac{dm_{i,t}^n}{dt}$ is the mass flow rate through heat node $i \in \psi_G \cup \psi_L$.

Dynamic Thermal Model

Consider a pipeline b segment whose length is Δx and cross-section is A . The following assumptions are introduced:

- Internal heat conduction is not considered. Then the enthalpy variation of the pipe segment is defined by the first law of thermodynamics, where the equation terms are in- and out-flow of enthalpy in longitudinal - horizontal direction and the radial heat loss flow in pipe segment [27].
- Pressure variation is too small to affect thermo-dynamic state. This implies that mass flow rate $\dot{m}$ does not change over horizontal axis x and specific enthalpy is equal to $C_p T$, where C_p is specific heat capacitance of the fluid and T is its temperature.
- The radial heat loss flow in a pipe segment can be simplified as $\lambda \Delta x (T - T^a)$, where T^a is ambient temperature and λ is overall heat transfer coefficient.

Applying the above assumptions to pipeline b and recognizing supply (superscript S) and return (superscript R) systems, the following PDEs are obtained [27]:

$$\frac{\partial \tau_{ij}^{S/R}}{\partial t} = - \frac{\dot{m}_{ij,t}^{S/R}}{\rho A_{ij}^{S/R}} \frac{\partial \tau_{ij}^{S/R}}{\partial x} - \frac{\lambda_{ij}^{S/R}}{\rho A_{ij}^{S/R} C_p} (\tau_{ij}^{S/R} - T_{i,t}^a) \quad (16)$$

where $\tau_{ij}^{S/R}$ are unknown temperatures and ρ is water density. In equation (16), it was assumed that ambient temperature varies with geographic location and time t .

Applying the finite difference method to approximate $\frac{\partial \tau_{ij}^{S/R}}{\partial t}$ and $\frac{\partial \tau_{ij}^{S/R}}{\partial x}$, the following equations are obtained:

$$\frac{\partial \tau_{ij}^{S/R}(x, t)}{\partial t} \approx \frac{\tau_{ij}^{S/R}(x, t) - \tau_{ij}^{S/R}(x, t - \Delta t)}{\Delta t} \quad (17)$$

$$\frac{\partial \tau_{ij}^{S/R}(x, t)}{\partial x} \approx \frac{\tau_{ij}^{S/R}(x, t) - \tau_{ij}^{S/R}(x - \Delta x, t)}{\Delta x} \quad (18)$$

Substituting equations (17) and (18) into equation (16), the heat dynamic process can be described as the bilinear difference equations:

$$\tau_{ij}^{S/R}(x, t) = \frac{\frac{\lambda_{ij}^{S/R} T_{i,t}^a}{\rho C_p A_{ij}^{S/R}} + \frac{\dot{m}_{ij,t}^{S/R}}{\Delta x \rho A_{ij}^{S/R}} \tau_{ij}^{S/R}(x - \Delta x, t) + \frac{1}{\Delta t} \tau_{ij}^{S/R}(x, t - \Delta t)}{\frac{1}{\Delta t} + \frac{\dot{m}_{ij,t}^{S/R}}{\Delta x \rho A_{ij}^{S/R}} + \frac{\lambda_{ij}^{S/R}}{\rho C_p A_{ij}^{S/R}}} \quad (19)$$

where $\Delta x = \frac{L_{ij}^{S/R}}{S_{ij}^{S/R}}$, and $S_{ij}^{S/R}$ is the number of segments (spatial points) along pipe ij in the supply/return side.

Moreover, the pipeline ij 's inlet temperature equals to the temperature of its connecting node i :

$$\tau_{ij}^{S/R}(0, t) = T_{i,t}^{S/R} \quad (20)$$

The temperature profile can be gradually calculated as illustrated in *Figure 6* [28]. For every time step Δt , $\tau_{ij}^{S/R}(x, t)$ for each pipe element is calculated using the previous temperature profile $\tau_{ij}^{S/R}(x, t - \Delta t)$ as initial condition and the pipe input temperature $\tau_{ij}^{S/R}(x - \Delta x, t)$ as boundary condition.

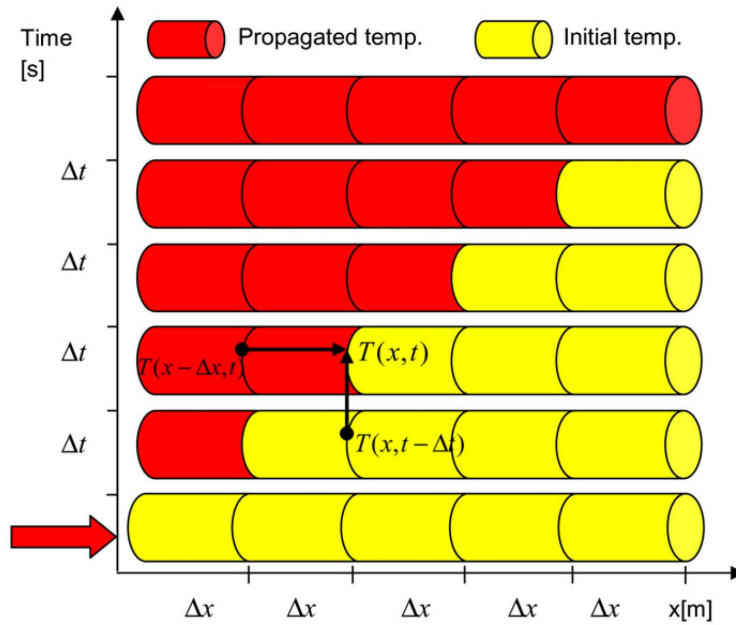


Figure 6: The thermal calculation.

To determine the network temperature profile at the end of the simulation time, the steps include:

- 1) The setting of an initial temperature of the different nodes in the network.
- 2) The calculation of the temperature propagation in each pipe after Δt sec using the finite difference method (equation (19)) and the boundary condition (equation (20)). In other words, through the temperature propagation, pipeline ij 's outlet temperature $\tau_{ij}^{S/R}(S_{ij}^{S/R}\Delta x, \Delta t)$ can be represented as a linear function of $T_{i,\Delta t}^{S/R}$. The detailed description of the calculation process is given in [28].
- 3) The calculation of the new node temperature after Δt using the nodal heat power constraint equation (7) and equations (10)–(15).
- 4) The repetition of the last two steps until the end of the simulation time.

3.4.3 Natural Gas Modeling

Natural gas flows are considered first under steady-state and then under transient conditions.

Static Gas Flow Model

Bernoulli's equation as given in equation (3) is considered and the following assumptions are introduced for the pipe flow modelling [21, 26]:

- Pipe cross sections A are constant, and the fluid is in a steady state, implying fluid velocity v is constant.
- Considering the gas continuity in pipes ($\rho v A = \text{const}$), the gas has constant fluid density ρ in pipe segments. This also means $\rho v = \text{const}$.
- For the constant fluid density ρ and velocity v , head loss due to friction in the pipe segment dx is defined by Darcy's equation [21]:

$$h_f = \frac{4f(\cdot)}{D} \frac{v^2}{2g} dx \quad (21)$$

where $f(\cdot)$ is the dimensionless friction factor and D is the internal diameter of the pipe. The friction coefficient $f(\cdot)$ is constant along the pipe segment length and calculated using different relations for laminar, partially turbulent, and turbulent regions.

- Heat transfer with the surrounding through the pipe walls can be neglected in the case of short distribution pipes and transmission pipe segments. This gives isothermal processes (i.e., constant temperature T) in gas pipes.
- Given the constant fluid temperature T and density ρ in pipes, the fluid compressibility factor Z , defined as $Z = \frac{p}{RT\rho}$, where R is a fluid constant related to the constant for air R_{air} , is therefore constant. Moreover, for standard temperature and pressure (STP) conditions ($p_{STP} \cong 0.1 \text{ MPa}$; $T_{STP} = 288 \text{ K}$), compressibility factor, $Z_{STP} = 1$.

Starting from equation (3), considering the above assumptions, integrating along a pipe from $x = 0$, $p = p_1$, to $x = L$, $p = p_2$, and assuming the pipe is horizontal (i.e., elevation term can be neglected), the general equation for steady-state volumetric flow q is obtained:

$$q_{STP} = \frac{\pi}{8} \times \sqrt{R_{air}} \times \frac{T_{STP}}{p_{STP}} \times \sqrt{\frac{(p_1^2 - p_2^2) \times D^5}{f(\cdot) \times SG \times L \times T \times Z}} \quad (22)$$

where SG is specific fluid gravity, or the ratio of substance mass to the air mass for the same unit of volume; pipe end nodes are denoted with subscripts "1" and "2", whilst subscript "STP" denotes quantities under STP conditions.

If a gas distribution system is supplied by a transmission network, the gas mix is the same in the entire network; for example, "natural gas" can have 95% of methane, 3.2% of ethane, and 1% of nitrogen [29]. On the other hand, in case there are alternative decentralized gas sources, variable percentages of gas components across the network are in place. The gas mix is modelled via molar fractions γ_u ; $u = 1, \dots, U_G$, so that mix specific gravity SG_{mix} and compressibility Z_{mix} must be replaced in equation (22) instead of SG and Z . The mix specific gravity is:

$$SG_{mix} = \frac{Z_{air}}{Z_{mix}} \times \frac{\sum_{u=1}^{U_G} (\gamma_u M_u)}{M_{air}} \quad (23)$$

where M_u is the molar mass of the component u , which gives steady-state flow q_{ij} between nodes i and j :

$$q_{ij} = \text{sgn}(p_i - p_j) \times \frac{\pi}{8} \times \sqrt{R_{air}} \times \frac{T_{STP}}{p_{STP}} \times \sqrt{\frac{M_{air} \times |p_i^2 - p_j^2| \times D^5}{f(\cdot) \times Z_{air} \times L \times T \times \sum_{u=1}^{U_G} (\gamma_u \times M_u)}} \quad (24)$$

The flow direction, from higher to lower pressure, is specified by function $\text{sgn}(p_i - p_j)$, while the friction coefficient $f(\cdot)$ is dependent on Reynolds number and flow type [25].

At each node i , the balance of volumetric gas flow equation (25) and gas component u in the gas mix equation (26) must be satisfied:

$$\sum_{b_i^{in}=1}^{B_i^{in}} q_{b_i^{in}} + q_{in/out_i} = \sum_{b_i^{out}=1}^{B_i^{out}} q_{b_i^{out}} , \quad i = 1, 2, \dots, N^G \quad (25)$$

$$\begin{aligned} & \sum_{b_i^{in}=1}^{B_i^{in}} \left(\gamma_{u_{b_i^{in}}} q_{b_i^{in}} \right) + \gamma_{u_{in/out_i}} q_{in/out_i} \\ & = \gamma_{u_{out_i}} \sum_{b_i^{out}=1}^{B_i^{out}} q_{b_i^{out}} , \quad u = 1, 2, \dots, U_G, i = 1, 2, \dots, N^G \end{aligned} \quad (26)$$

where b_i^{in} and b_i^{out} are indices of branches with gas flowing in and out of node i , respectively, B_i^{in} and B_i^{out} denote the numbers of branches with gas flowing in and out of node i , respectively, q_{in/out_i} is the gas injection or offtake at node i , $\gamma_{u_{b_i^{in}}}$ represents a flow-in molar fraction of gas component u in branch b_i^{in} , and $\gamma_{u_{out_i}}$ is component u molar fraction after a perfect gas mix at node i ; and N^G is the number of nodes in the gas network.

Dynamic Gas Flow Model

Considering a gas pipeline and introducing the following assumptions:

- A rigid pipeline b has a constant diameter D_b and cross-section A_b . The pipeline is horizontal and there is no elevation (gravitational) term in the momentum equation.
- Pipeline b gas temperature T_b is constant, the energy equation is not needed, and temperature is not a state [30].
- The compressibility factor Z_b along pipeline b is constant.
- Assumptions 2) and 3) give a linear relationship between pipe pressure p_b and gas density ρ_b in the state equation $p_b = Z_b R T_b \rho_b = c^2 \rho_b$. This means they can be interchangeably used.
- Since pipeline b gas velocity v_b is much smaller than the speed of sound c , the convective term $\frac{\partial(\rho_b v_b^2)}{\partial t}$ in the momentum equation can be neglected. Similarly, acceleration (inertial) term $\frac{\partial(\rho_b v_b)}{\partial t}$ in the momentum equation can also be neglected [31].
- The selected states in the pipe b model is mass flow rate $f l_b$ and pressure p_b .

The gas flow dynamics are generally described by mass conservation equation, momentum equation and energy equation. Considering the above assumptions, the mass conservation and momentum equation at point (x, t) can be written in the form:

$$\frac{\partial f l_b}{\partial x} + K_1 \frac{\partial p_b}{\partial t} = 0 \quad (27)$$

$$K_2 p_b \frac{\partial p_b}{\partial x} + f l_b |f l_b| = 0 \quad (28)$$

where $f l_b$ and p_b are gas flow mass rate and pressure in pipe b , respectively; x and t represent space and time dimensions, respectively; and K_1 and K_2 are constant parameters of the transient equations, respectively [32]. Note that “higher level” constants K_1 and K_2 contain parameters described in equation (24).

Equations (27) and (28) can be numerically solved to obtain temporal and spatial variations (i.e., dynamics) of the gas flow and pressure in pipe b . Several groups of methods, such as the finite difference method, the method of lines, the finite element method, the finite volume method, and/or the gradient discretization method, can be used for this purpose [33]. It was assumed, for simplicity, that the time step $\Delta t = 1 \text{ h}$ and the spatial step $\Delta x = L_{ij}$. Then, by using the implicit finite difference method [34], equations (27) and (28) are transformed into algebraic equations:

$$\frac{(f l_{ij,t}^{\text{inlet}} - f l_{ij,t}^{\text{outlet}})}{L_{ij}} + \frac{K_1 \left(\frac{p_{i,t} + p_{j,t}}{2} - \frac{p_{i,t-1} + p_{j,t-1}}{2} \right)}{1} = 0 \quad \forall (ij) \in G_b, t \in G_t \quad (29)$$

$$\frac{K_2 (\pi_{i,t}^2 - \pi_{j,t}^2)}{L_{ij}} + \bar{f} l_{ij,t} |\bar{f} l_{ij,t}| = 0 \quad \forall (ij) \in G_b, t \in G_t \quad (30)$$

where $f l_{ij,t}^{\text{inlet}}$ and $f l_{ij,t}^{\text{outlet}}$ are, respectively, inlet and outlet flow rate in the pipeline ij ; L_{ij} is the length of the pipeline ij ; G_b is the set of branches; G_t is the set of time intervals; and $\bar{f} l_{ij,t} = \frac{f l_{ij,t}^{\text{inlet}} + f l_{ij,t}^{\text{outlet}}}{2}$ is the average flow rate in pipe ij . Further, define the modified line-pack and Weymouth constant of the pipeline ij in period t :

$$LP_{ij,t} = \frac{p_{i,t} + p_{j,t}}{2} L_{ij} \quad \forall (ij) \in G_b, t \in G_t \quad (31)$$

$$WC_{ij} = \sqrt{\frac{K_2}{L_{ij}}} \quad \forall (ij) \in G_b, t \in G_t \quad (32)$$

so that equations (31) and (32) are equivalent to:

$$f l_{ij,t}^{\text{inlet}} - f l_{ij,t}^{\text{outlet}} = K_1 (LP_{ij,t} - LP_{ij,t-1}) \quad \forall (ij) \in G_b, t \in G_t \quad (33)$$

$$\frac{\bar{f} l_{ij,t} |\bar{f} l_{ij,t}|}{WC_{ij}^2} = p_{i,t}^2 - p_{j,t}^2 \quad \forall (ij) \in G_b, t \in G_t \quad (34)$$

Equations (33) and (34) constitute a quasi-steady-state model in each hourly interval that considers the line pack (i.e., the natural gas stored in each pipeline).

Model equations (30)-(31) (or equations (33)-(34)) gives discretized non-linear algebraic equations for each branch $(ij) \in G_b$. When modelling the entire gas network, balances

of mass flow rates at each node must be satisfied. Assuming there is only one gas component and there are gas compressors beside pipelines, the balance equation at each node i is:

$$\sum_{s \in G_s(i)} fl_{s,t}^S - \sum_{d \in G_d(i)} fl_{d,t}^D - \sum_{k \in G_C(i)} \varphi_{k,t} = \sum_{j \in G_b(i)} fl_{ij,t}^{inlet} + \sum_{k \in G_C(i)} fl_{k,t}^C \quad \forall i \in G_n, t \in G_t. \quad (35)$$

where $fl_{s,t}^S$ is gas production at node s connected with node i ; $fl_{d,t}^D$ is gas demand at node d ; $\varphi_{k,t}$ is fuel consumption of gas-driven compressor at node k ; $fl_{ij,t}^{inlet}$ is inlet gas flow in pipeline ij ; $fl_{k,t}^C$ is compressor outflow towards node k ; G_s , G_d , G_C , and G_n are sets of source nodes, demand nodes, compressors and all nodes, respectively. The fuel consumption of natural gas-driven compressors is computed by:

$$\varphi_{k,t} = \vartheta_k fl_{k,t}^C \quad \forall k \in G_C, t \in G_t. \quad (36)$$

where ϑ_k is the conversion efficiency of the natural gas compressor k .

3.4.4 Energy Circuit Method for Heterogeneous Energy System

The traditional modeling methods for each system of the electric, heat, and gas network are independent of each other and have obvious differences, which brings obstacles to the unified analysis of the IESSs. Besides, the dynamic analysis of large-scale IESSs takes a great mass of time. It has a numerical convergence problem, which hinders the application of the Integrated Energy Management System (IEMS) including the IES-SE in large-scale IESSs. The energy circuit method is a novel circuit-analog modeling method for IESSs that fully draws on the deduction of electric circuit modeling from the time domain to the frequency domain and from distributed parameters to lump parameters. This method models the gas and heat networks with lumped parameters in the frequency domain, which can unify the analysis process of heterogeneous energy flow networks such as electric, heat, and gas in the same mathematical form. It effectively algebraizes the original PDE model of dynamic processes in natural gas networks and thermal networks, which avoids finite differences in time and space, thus reducing the computational burden and accelerating the solution speed more than hundreds and thousands of times. It also has the advantages of high parallel computation efficiency and no numerical convergence problem. The energy circuit method is especially suitable for large-scale IESSs, making the online and real-time management and control of large-scale IESSs considering dynamic processes possible, which is a great step forward in modeling heterogeneous energy systems. It provides a unified and solid theoretical foundation of developing and applying IEMS [20, 35-38].

The basic steps of the energy-circuit method can be divided into branch-oriented modeling and network-oriented modeling.

a) Energy Circuit Modeling for Pipelines

The voltage and current along AC transmission lines satisfy the telegrapher's equations, which are PDEs corresponding to a time domain distributed parameter (TDDP) electric circuit model. Using the phasor method that converts a sine wave in the time domain into a complex vector to eliminate the temporal derivative, these PDEs are simplified to ODEs, corresponding to a frequency domain distributed parameter (FDDP) electric circuit model. Further solving these ODEs yields an algebraic

relationship between the voltage and current at both ends, which corresponds to a frequency domain lump parameter electric circuit model, i.e., the famous π -equivalent electric circuit [39]. Such a two-step transformation has brought significant computational convenience for power system analysis.

By analogizing the circuits in the following table, PDEs describing the relationship between pressure and gas flow in natural gas pipelines and the relationship between temperature and heat flow in heat networks can be derived.

Table 2: Circuit Analogy from Electric Circuit to Fluid and Thermal Circuit

Energy	Electric Circuit	Fluid Circuit	Thermal Circuit
Physical Object	Transmission lines in electricity networks	Natural gas pipelines in natural gas networks	Heating pipelines in heating networks
Potential P	Voltage U	Pressure p	Temperature T
Flow F	Current I	Gas flow m	Heat flow h
Resistance resists flow	$R_e: U = R_e I$	$R_f: p = R_f m$	$R_t: T = R_t h$
Conductance resists potential	$G_e: I = G_e U$	$G_f: m = G_f p$	$G_t: h = G_t T$
Inductance resists flow change	$L_e: U = L_e dI/dt$	$L_f: p = L_f dm/dt$	$L_t: T = L_t dh/dt$
Capacitance resists potential change	$C_e: I = C_e dU/dt$	$C_f: m = C_f dp/dt$	$C_t: h = C_t dT/dt$

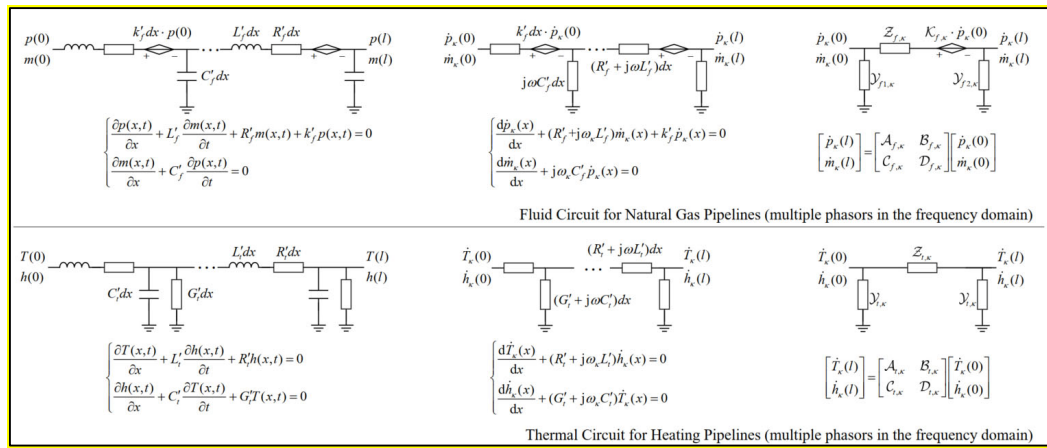


Figure 7: Energy circuit modeling from time domain to frequency domain, and from partial differential equations to algebraic equations on mathematics.

As shown in *Figure 7*, the system of PDEs describing the gas and heat networks can be rewritten in the form of telegraph equations, thus realizing a two-step conversion of the transmission line circuit model and deriving computationally efficient energy circuit models for natural gas pipelines and heating pipelines.

$$\begin{aligned}
 A \frac{\partial p}{\partial t} + c^2 \frac{\partial m}{\partial x} &= 0 \\
 \frac{1}{A} \frac{\partial m}{\partial t} + \frac{\partial p}{\partial x} + \frac{\lambda \rho v^2}{2D} + \rho g \sin \alpha &= 0 \\
 m &= \rho v A \\
 p &= c^2 \rho
 \end{aligned} \tag{37}$$

where constants A , D , λ , α , g and c are the cross-sectional area, inner diameter, friction factor, dip angle, gravitational acceleration, and sonic speed, respectively; variables p , m , v , and ρ are the pressure, mass flow rate, velocity, and density, respectively.

$$\begin{aligned}
 c_p \rho A \frac{\partial T}{\partial t} + c_p m \frac{\partial T}{\partial x} + \mu T &= 0 \\
 h &= c_p m T
 \end{aligned} \tag{38}$$

where constants c_p and μ are the specific heat capacity and heat dissipation coefficient, respectively, and variables T and h are the relative temperature and heat flow, respectively. In this section, only heating networks operated with the quality regulation mode are considered, in which the water flow m of pipelines is a known constant optimized in advance for typical operating conditions and the outlet temperature of heating sources is adjusted to follow heat load changes.

More details about the circuit analogies are provided in *Table 2* and the energy circuit modeling in *Figure 7* is explained as follows.

Circuit Analogy

As shown in *Table 2*, state variables of the different energy flows are divided into two categories i.e., potential and flow. The potential variables include the voltage U in the power network, the pressure p in the natural gas network, and the temperature T in the heating network; and the flow variables include the current I in the power network, the gas flow m in the natural gas network, and the heat flow h in the heating network. Based on this, the current resistor R , the conductance G of the resisting potential, the current resisting varying inductance L and capacitance C varying in resistive potential are generalized from circuits (subscripted as c) to fluid circuits (subscripted as f) and thermal circuits (subscripted as t). By extracting these circuit components, the PDEs for the natural gas pipeline and the heating pipeline in equations (37) and (38) are reorganized:

$$\begin{aligned}
 \frac{\partial p(x, t)}{\partial x} + L'_f \frac{\partial m(x, t)}{\partial t} + R'_f m(x, t) + k'_f p(x, t) &= 0 \\
 \frac{\partial m(x, t)}{\partial x} + C'_f \frac{\partial p(x, t)}{\partial t} &= 0
 \end{aligned} \tag{39}$$

$$\begin{aligned}
\frac{\partial T(x, t)}{\partial x} + L'_t \frac{\partial h(x, t)}{\partial t} + R'_t h(x, t) &= 0 \\
\frac{\partial h(x, t)}{\partial x} + C'_t \frac{\partial T(x, t)}{\partial t} + G'_t T(x, t) &= 0
\end{aligned} \tag{40}$$

These two PDEs give the TDDP circuits of natural gas pipelines and heating pipelines, as shown in the first column of *Figure 7*. The expressions of the involved circuit component parameters including R'_f , L'_f , k'_f and C'_f are provided below:

$$\begin{aligned}
R'_f &= \lambda v_b / AD \\
L'_f &= \frac{1}{A} \\
C'_f &= \frac{A}{c^2} \\
k'_f &= (2Dg \sin \alpha - \lambda v_b^2) / 2Dc^2
\end{aligned} \tag{41}$$

Fourier Transform and Phasor Representation

To enable the phasor method, the Fourier transform is employed to decompose non-sinusoidal excitations in natural gas networks and heating networks into several sinusoidal components with different frequencies. The desired response is exactly the time-domain superposition of responses for each frequency component, as visualized in *Figure 8*.

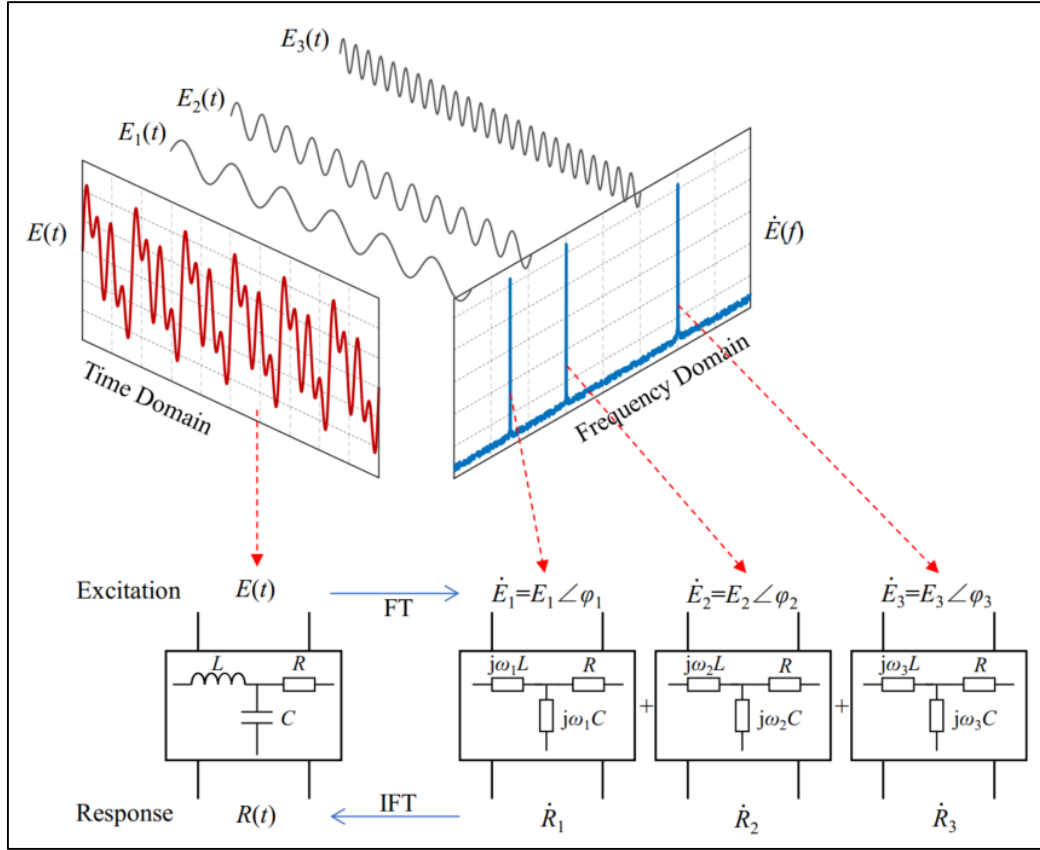


Figure 8: The decomposition of excitations and the recovery of responses.

For k th frequency component whose angular frequency is ω_k , the TDDP equations in (39) and (40) are rewritten using the phasor representation:

$$\begin{aligned} \frac{d\dot{p}_k(x)}{dx} + (R'_f + j\omega_k L'_f)\dot{m}_k(x) + k'_f \dot{p}_k(x) &= 0 \\ \frac{d\dot{m}_k(x)}{dx} + j\omega_k C'_f \dot{p}_k(x) &= 0 \end{aligned} \quad (42)$$

$$\begin{aligned} \frac{d\dot{T}_k(x)}{dx} + (R'_t + j\omega_k L'_t)\dot{h}_k(x) &= 0 \\ \frac{d\dot{h}_k(x)}{dx} + (G'_t + j\omega_k C'_t)\dot{T}_k(x) &= 0 \end{aligned} \quad (43)$$

These two ODEs give FDDP circuits for natural gas pipelines and heating pipelines, respectively. Unlike the circuits, the TDDP circuits for fluid and heat circuits correspond to a series of FDDP circuits at different frequencies.

Two-Port Equivalence

Given the boundary values at the beginnings of pipelines, the FDDP equations of k th frequency component in equations (42) and (43) are analytically solved, whose solutions are as follows:

$$\begin{bmatrix} \dot{p}_k(l) \\ \dot{m}_k(l) \end{bmatrix} = \begin{bmatrix} A_{f,k} & B_{f,k} \\ C_{f,k} & D_{f,k} \end{bmatrix} \begin{bmatrix} \dot{p}_k(0) \\ \dot{m}_k(0) \end{bmatrix} \quad (44)$$

$$\begin{bmatrix} \dot{T}_k(l) \\ \dot{h}_k(l) \end{bmatrix} = \begin{bmatrix} A_{t,k} & B_{t,k} \\ C_{t,k} & D_{t,k} \end{bmatrix} \begin{bmatrix} \dot{T}_k(0) \\ \dot{h}_k(0) \end{bmatrix} \quad (45)$$

b) Energy Circuit Modeling for Networks

As mentioned before, an energy network with non-sinusoidal excitations can be analyzed by converting original excitations into multiple sinusoidal components with different frequencies. Thus, the energy circuit modeling for networks under a single sinusoidal excitation component is focused in this subsection, which is derived as follows. Note the subscript k that distinguishes different frequency components is omitted for the purpose of brevity.

Branch Equation

According to the π -equivalent energy circuit models, each transmission line in electricity networks or pipeline in natural gas networks and heating networks can be modeled as three branches that are composed of impedance and controlled sources. Thus, a unified branch equation of these networks is given in matrix form as follows:

$$\dot{F}_b = Y_b(\dot{P}_b - K_b \dot{P}_{bf}) \quad (46)$$

where, $\dot{F}_b$ is a vector whose i th entry is the flow variable of i th branch, $\dot{P}_b$ is a vector whose i th entry is the potential difference of i th branch, $\dot{P}_{bf}$ is a vector whose i th entry is the potential variable on the from side of i th branch, Y_b is a diagonal matrix whose i th entry is the admittance of i th branch, and K_b is a diagonal matrix whose i th entry is the pressure-controlled pressure source parameter of i th branch. Particularly, K_b is a zero matrix for electricity networks and heating networks.

Topology Equation

The node-branch incidence matrix A and node-outflow-branch incidence matrix A_+ are introduced to indicate the network topology, whose definitions are given below.

$$A(i, j) = \begin{cases} 1, & \text{if branch } j \text{ flows from node } i \\ -1, & \text{if branch } j \text{ flows into node } i \\ 0, & \text{otherwise} \end{cases} \quad (47)$$

$$A_+(i, j) = \begin{cases} 1, & \text{if branch } j \text{ flows from node } i \\ 0, & \text{otherwise} \end{cases} \quad (48)$$

Where, $A(i, j)$ and $A_+(i, j)$ are their entries at the i th row and j th column, respectively.

Based on these two matrices, the KCL and KVL constraints in electricity networks are generalized to the flow constraint and potential constraint. The flow constraint means that the difference between the outflow and inflow of a node equals its flow injection, as formulated in equation (49). The potential constraint correlates the branch potential and node potential, as formulated in equations (50) and (51).

$$A\dot{F}_b = \dot{F}_n \quad (49)$$

$$A^T \dot{P}_n = \dot{P}_b \quad (50)$$

$$A_+^T \dot{P}_n = \dot{P}_{bf} \quad (51)$$

where, $\dot{F}_n$ is a vector whose i th entry is the flow injection of i th node and $\dot{P}_n$ is a vector whose i th entry is the potential variable of i th node.

Network Equation

Substituting the topology equations (49)-(51) into the branch equation (46) yields a unified network equation for electricity networks, natural gas networks, and heating networks, as in equation (52).

$$\dot{F}_n = Y_n \dot{P}_n \quad (52)$$

where $Y_n = AY_b A^T - AY_b K_b A_+^T$ is named the generalized node admittance matrix.

Thus far, the intractable PDEs that model natural gas networks and heating networks under general excitations have been equivalently transformed into a series of algebraic equations in the form of equation (52). Because this algebraization method is generalized from the electric circuit method (ECM), it can be called ECM. Compared with the FDM that achieves the same function, the proposed ECM produces fewer variables and equations, which thereby yields better computation efficiency. This performance improvement is mainly attributed to avoiding spatial discretization by two-port equivalence in the ECM. In addition, different frequency components in the ECM are independent and therefore can be calculated in a parallel manner, while different time steps in the FDM have to be calculated serially. This distinction further widens the performance gap between the two methods.

3.4.5 Modeling Differences between Integrated Energy Systems - State Estimation and Traditional Tools

- a) The filtering performance of IES-SE methodology is different from that of the traditional Power System - State Estimation (PS-SE) methodology. The main significance of IES-SE methodology is to improve the redundancy of the system by coupling constraints, obtain a globally consistent solution with higher accuracy, and realize processed bad data at the boundaries of different energy systems.
- b) The monitoring level and communication infrastructure of heat and gas networks are much lower than that of electrical networks for technical and economic reasons, and the probability of SE being affected by bad data is much higher than that of electrical networks, so IES-SE methodology requires the higher performance of robustness, convergence, and communication than PS-SE.
- c) In most cases, it is difficult to realize the regulation of different energy systems in IESs by unified EMS, so the IES-SE methodology must consider reliable SE under limited data interaction [40-43].
- d) When there is a disturbance in IESs, the time scale of measurements [22] and dynamic responses are inconsistent between the thermal, natural gas, and power systems, requiring new methodologies to coordinate.

At present, research on the physical models of the power system has been relatively mature, while the physical models of the gas network and heat network still face difficulties in application. For example, to improve the accuracy, the time step and the spatial step of the finite difference method should be reduced, which increases the computational burden. Thus, IES-SE methodology needs to balance the accuracy and the computational burden posed by the physical models and become more practical.

3.4.6 Coupling Device Modeling

There are coupling components between different subsystems of IES. For example, CHP units generate electricity and heat simultaneously. In this regard, this subsection covers coupling device modeling in IES, as follows:

Gas turbines and internal combustion engines are modeled as:

$$c_{m,i} = \frac{\Phi_i^{gt}}{P_i^{gt}} \quad (53)$$

where Φ_i^{gt} and P_i^{gt} are the thermal power production and electric power production of the gas turbine, respectively, and $c_{m,i}$ is a constant ratio.

When operating in the condensing mode, the extraction steam turbine can be modeled as:

$$\eta_{s,i} = \frac{\Delta\Phi_i^{st}}{\Delta P_{scr,i}^{st}} = \frac{\Phi_i^{st}}{P_{cap,i} - P_i^{st}} \quad (54)$$

$$P_{cap,i} = \eta_e G_{in,i} \quad (55)$$

where Φ_i^{st} and P_i^{st} are the thermal power production and electric power production of the steam turbine, respectively; $\eta_{s,i}$ is a heat-electricity production ratio; $P_{cap,i}$ is the maximum electric power production; $G_{in,i}$ is the power of the steam; and η_e is the conversion ratio.

The circulating pump needs power supply from the electric power system. The circulating pump with efficiency of η_i^{pu} needs to consume power:

$$P_{pu,i} = \frac{m_i h_i}{g \times \eta_i} \quad (56)$$

where, m_i is the mass flow; h_i is the pressure difference; and η_i is the efficiency.

The heat pump consumes electricity to transfer heat from a cool space to a warm space. The heat-power relation can be expressed as:

$$\eta_{hp} = \frac{\phi_{hp}}{P_{hp}} \quad (57)$$

where η_{hp} is the efficiency of the heat pump; ϕ_{hp} is the transferred heat; and P_{hp} is the consumed electric power.

The electric boiler is another type of devices for heat-power conversion:

$$\eta_b = \frac{\phi_b}{P_b} \quad (58)$$

where η_b is the efficiency of the electric boiler; ϕ_b is the produced heat; and P_b is the consumed electric power.

Besides the widely deployed coupling devices described above, there are many emerging types of coupling devices in IES, which require proper modeling. Some of the examples include power-to-gas technologies, fuel cells, electric vehicle chargers, etc.

3.5 Necessity of Integrated Energy Systems-Energy Management System

In the traditional electricity, heating/cooling, gas, and other fields, there are relatively mature monitoring and management techniques, but they have not been unified. Along with the IES development, in order to achieve a comprehensive, real-time, and accurate perception of IES, IES-SE is needed. It can provide trusted data for a multi-energy stream EMS and obtain the optimally estimated value of the IES state variable. Only on this basis, the unified management and optimal scheduling of multi-energy flow can be realized as shown in *Figure 9*. In this sense, the multi-energy flow-oriented IES-SE is the foundation and core of the IEMS. In the field of power systems, EMS, first proposed by Dyalacco in 1967, has become relatively mature after half a century of development. EMS uses real-time data from Supervisory Control and Data Acquisition (SCADA) and wide-area measurement systems to perform closed-loop (or semi-closed-loop) controls. In order to realize the security, reliability, resiliency, and economy of the power system operation, EMS is recognized as the "central brain" of the power system operation. However, the traditional EMS is limited to the management and scheduling the electrical energy in power systems. The physical characteristics, coupling mechanism, and modeling methods of multi-energy flows are essentially different from those of pure electrical energy flows, and the traditional EMS cannot be directly applied to IES. To enhance the operational performance of IES, it is urgent to propose and develop the multi-energy flow-oriented IEMS [6].

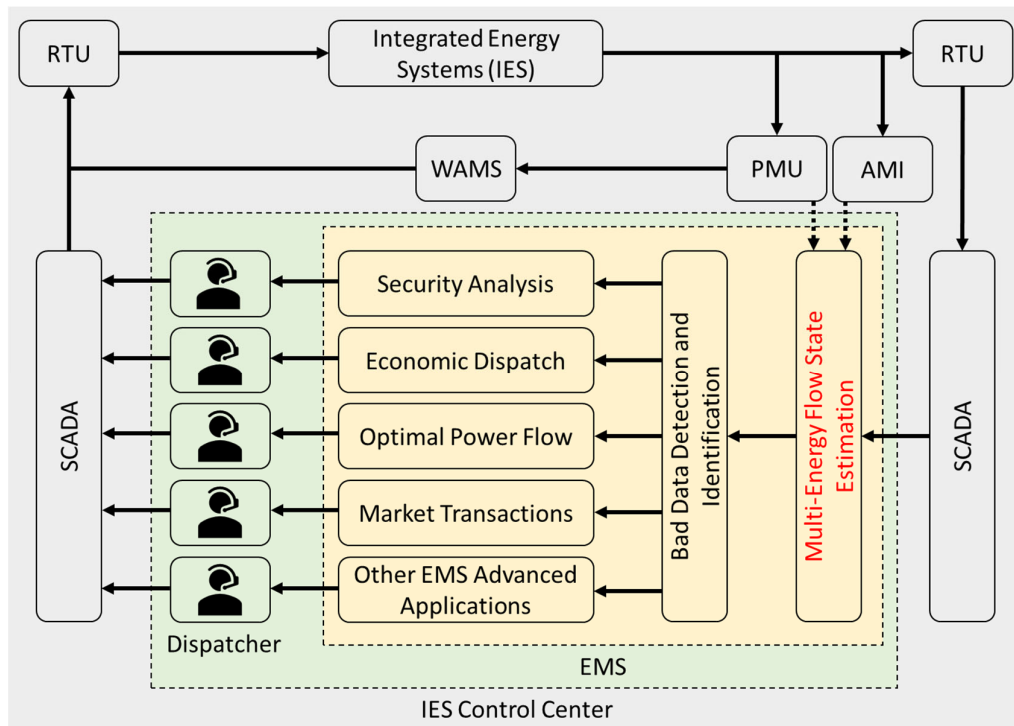


Figure 9: Schematic diagram of EMS for IES.

4. State Estimation of IESs

A brief introduction to SE for IESs is presented in the following subsections.

4.1 Motivations for Integrated Energy Systems-State Estimation

SE not only estimates and predicts the operating state of the system but also performs the function of data filtering. SE can estimate unmeasured quantities, suppress measurement noise and bad data, evaluate the rationality of the model and the accuracy of the system measurement instruments. In addition to the above advantages, IES-SE can make full use of the advantage of high measurement redundancy of power systems to improve the estimation accuracy of the natural gas systems and thermal systems through coupling constraints. Besides, PDEs for natural gas and thermal systems can also provide additional power systems constraints at the coupling points. The high-performance multi-energy stream IES-SE can provide accurate and reliable real-time operation data for IEMS, which enables and safeguards the operation of all advanced applications of IEMS (such as optimal generation controls, optimal energy flows, security assessment, generation and load forecasting, post-disaster system restoration, etc.). At present, with the rapid development of IES, it is urgent to accelerate the research related to IES-SE for multi-energy flows.

4.2 Similarities and Differences between Integrated Energy Systems-State Estimation and Power System-State Estimation

- a) **Both include Static State Estimation (SSE) and Dynamic State Estimation (DSE):** PS-SE includes both static and dynamic methods. When IES is in a steady state or quasi-steady state, it can be modeled by Integrated Energy Systems - Static State Estimation (IES-SSE). At this time, each system's characteristics and coupling equipment that compose IESs are described by algebraic equations. DSE involves the generator's dynamic process, which is described by ordinary

differential equations [44, 45]. The estimation results of IES-SSE can be used as the initial value of Integrated Energy Systems - Dynamic State Estimation (IES-DSE). Due to the different time scales of the systems that comprise IES, the application range of IES-DSE is broader than that of IES-SSE. In actual operation, the operating time scale of dynamics of the power systems is from milliseconds to minutes. In contrast, the operating time scale of the natural gas systems ranges from minutes to hours and the operating time scale of the thermal systems ranges from hours to days. Therefore, the operating time scale of the power systems is usually taken as the benchmark in the modeling of IES-DSE. The power systems can be regarded as in a steady state and described by algebraic equations. However, the natural gas systems and the thermal systems are described by partial differential equations [46-48]. As IES-DSE generally involves both systems characterized by algebraic equations and differential equations, the integration of SSE and DSE methods must be investigated [49, 50].

- b) **Both include centralized state estimation and decentralized state estimation:** PS-SE includes both centralized and decentralized estimation methods. The latter can improve reliability, computational efficiency, and data privacy compared to the centralized state estimation methods. However, bad data at boundaries and coordination between different systems in IESs must be carefully handled by decentralized estimation paradigms. Based on the characteristics of IES, IES-SE also includes centralized and decentralized estimation methods. There is generally a centralized energy management center for a uniformly managed IES, which can obtain the information of each subsystem of IES and implement centralized IES-SE with a relatively simple program design. But when IES does not have a centralized management center, it is infeasible to exchange all data in real-time in order to due to data security and privacy concerns. In this case, the centralized IES-SE method is not applicable, and the decentralized IES-SE method should be adopted.
- c) **Different state variables, measurements, measurement equations, and state transition equations:** The state variables of PS-SE are node voltage phasors (SSE) and generator dynamic variables (DSE). Measurements usually include node voltages, node injected powers, branch power flows, and dynamic variables of generators, etc. In addition to the node voltage phasors of the power systems, the state variables in IES-SE also include the pressures in the natural gas systems, and the pressures (or flow rates), and temperatures in the thermal systems. The IES-SE measurement must include the pressures and flow rates in the natural gas systems and the pressures, flow rates, and temperatures in the thermal systems, as well as the electrical measurements of the power systems. Under transient conditions (IES-DSE), the state of charge of batteries, the amount of hot water stored, and the amount of gas stored, might also be needed to initialize the process and to establish boundary constraints.

Different state variables and measurements lead to different forms of measurement equations for PS-SE and IES-SE. In addition to the measurement equation, the natural gas systems and the thermal systems that constitute IES also include the state transition equations. Generally, the state transition

equations of the thermal systems can be expressed by three categories of PDEs: momentum equations, mass balance equations, and energy equations [24-26]. The state transition equations of the natural gas systems can also be expressed by three equations: momentum equations (PDE), material balance equations (PDE), and state equations (algebraic equations) [24-26].

- d) Consideration of coupling devices in IES-SE:** In a power system, the equipment is homogeneous, so PS-SE does not need to address problems related to coupling equipment. IES contains a variety of coupled devices with different physics. Therefore, in the IES-SE model, it is necessary to associate multiple types of heterogeneous energy flow models in different systems with the coupled device models. The physics of coupling devices also introduces constraints serving as virtual measurements in IES-SE, enhancing the observability for state variables and the information redundancy for processing bad data. Thus, coupling devices must be considered in the IES-SE model.

The above comparison between PS-SE and IES-SE is summarized in *Table 3*. Clearly, the models and methods of IES-SE and PS-SE are different, and IES-SE needs detailed studies as an emerging topic and technology.

Table 3: Comparison of PS-SE and IES-SE

Objects	PS-SE	IES-SE
SE method	Both include SSE and DSE	
State variables	Node voltage vector and generator dynamic variables	State variables in the power systems, pressure (or flow rate) and temperature in the thermal systems, pressure (or flow rate) in the gas system
Measurement	Generator dynamic variable, node voltage amplitude, node voltage phase angle, node injected power (including active and reactive power), Branch power	Measurements in the power systems, pressure and flow in the gas systems, and pressure, flow, and temperature in the thermal systems
Measure equation	Algebraic equation	
State transition equation	Ordinary differential equation	Partial differential equation
Coupling unit	No coupling unit	Multiple coupling units
Implementation and architectures	Mainly centralized SE	Decentralized SE

4.3 Integrated Energy Systems-State Estimation Models (Multi-time Scales)

4.3.1 Actual Value vs Per-unit Value

Electric power systems are typically composed of different voltage zones with quantities with different orders of magnitude. This problem is solved by expressing voltage magnitudes in per unit, which makes them of the same order of magnitude. Voltage angles are usually given in radians or degrees. On the other hand, state variables of heat and gas systems can be expressed in both per units and actual value. Heat system states are expressed in per unit [23, 51]. This helps avoid ill-conditioning of the gain matrix caused by discrepant orders of measurement weights. The electricity network and district heating network are coupled by power, so the reference (base) values of electrical power and heat power should be equal when converting integrated electricity and heat system into per-units. However, highly non-linear measurement equations in the case of the gas system, especially when gas molar fractions are modeled for multi-component gas mixtures, make the transformation into the per unit domain very difficult. If heat system states are also given in actual values, this results in the multi-vector composed of a variety of states with different orders of magnitude. This can further introduce numerical problems into IES-SE, such as ill-conditioning or local optimum. Another problem, in this case, is how to evaluate the estimation accuracy over the widespread multi-vector; usually, accuracy must be evaluated over each type of state variable, which makes results interpretation challenging.

4.3.2 State Variables in State Estimation of IESs

Algebraic state variables in power systems are voltage magnitudes and phase angles (except for the reference node). In district heating systems, state variables describing both hydraulic and thermal phenomena must be specified. However, the selection of state variables is not unique such as in power systems. Algebraic state variables in heat systems are usually mass flow rates along the branches and temperatures at the nodes of the network [22, 52]. Some other works in [23, 51, 53, 54] report the use of nodal pressures/branch head drops together with nodal temperatures. In gas systems, several approaches are applied. For example, squared gas nodal pressures are used as states in [52], gas nodal pressures in [41, 55], and nodal pressures and pipeline flow rates are used as states in [56, 57]. The nodal gas densities and mass flow rates at both ends of pipelines are taken as state variables in [58, 59]. The auxiliary state variables can also be introduced into multi-vector to obtain a linear state estimation model [53]. Dynamic state variables of power systems can be internal states of machines or dynamic loads. Time constants of dynamic processes within power systems can vary over a wide range. If the power system dynamic process has much lower time constants as compared to those of the heat and/or gas dynamic process of interest, which is usually the case, the state variables of the power system are only algebraic variables and SE model for power system can be described via algebraic measurement function.

4.3.3 State Estimation Models for IESs: High-level Framework

Most of the reported SE models for IES are developed starting from the measurement model of IES:

$$\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{v} \quad (59)$$

where measurement vector $\mathbf{z}$ is composed of electricity, gas, and heat measurements $\mathbf{z} = [\mathbf{z}_e^T, \mathbf{z}_g^T, \mathbf{z}_h^T]^T$, $\mathbf{h}$ is vector-valued measurement function, state vector $\mathbf{x}$ is composed of electricity, gas and heat states $\mathbf{x} = [\mathbf{x}_e^T, \mathbf{x}_g^T, \mathbf{x}_h^T]^T$, and $\mathbf{v}$ is the measurement noise vector assumed to be normally distributed with zero mean and covariance matrix $\mathbf{R}$. In this case, IES-SE model is a sum of three objectives, where each objective is the weighted sum of squared differences between actual and estimated measurements [22, 23, 52-54, 57, 60-62]. The state estimation of IES, which is composed of three main energy systems, seeks to solve the following Constrained Weighted Least Square (CWLS) optimization:

$$\begin{aligned} \min J(\mathbf{x}_e, \mathbf{x}_g, \mathbf{x}_h) \\ = [\mathbf{z}_e - \mathbf{h}_e(\mathbf{x}_e)]^T \mathbf{R}_e^{-1} [\mathbf{z}_e - \mathbf{h}_e(\mathbf{x}_e)] + [\mathbf{z}_g - \mathbf{h}_g(\mathbf{x}_g)]^T \mathbf{R}_g^{-1} [\mathbf{z}_g - \mathbf{h}_g(\mathbf{x}_g)] \\ + [\mathbf{z}_h - \mathbf{h}_h(\mathbf{x}_h)]^T \mathbf{R}_h^{-1} [\mathbf{z}_h - \mathbf{h}_h(\mathbf{x}_h)] \end{aligned} \quad (60)$$

s. t. $\mathbf{c}(\mathbf{x}) = \mathbf{0}$

where $\mathbf{x}_j$ is the state vector of j^{th} energy system, $\mathbf{z}_j$ is the measurement vector of j^{th} energy system, $\mathbf{h}_j$ is the measurement function of j^{th} energy system, $\mathbf{R}_j$ is measurement noise covariance matrix of j^{th} energy system and $\mathbf{c}(\mathbf{x})$ is a set of constraints in all energy systems (including topological constraints and coupling constraints). For example, a two-stage approach [22] is used to estimate heat flow delays in pipes in the first stage and then all electricity and heat states via CWLS in the second stage. There are also applications in which the CWLS method is used to obtain estimates within a (moving) time window. In [62], initial PDEs describing the gas system are linearized first and then converted into algebraic equations in the frequency domain giving the gaseous circuit method.

The CWLS method is a non-convex optimization problem with well-known deficiencies. One possibility to overcome difficulties is to replace the nonlinear non-convex measurement model with a convex bilinear model [52]. Yet another possibility is to “convexify” Lagrangian by adding a quadratic penalty term:

$$\min L^A(\mathbf{x}_e, \mathbf{x}_g, \mathbf{x}_h, \boldsymbol{\lambda}) = J(\mathbf{x}_e, \mathbf{x}_g, \mathbf{x}_h) + \boldsymbol{\lambda}^T \mathbf{c}(\mathbf{x}) + \frac{1}{2} \rho \|\mathbf{c}(\mathbf{x})\|_2^2 \quad (61)$$

where $\boldsymbol{\lambda}$ is a vector of Lagrange multipliers and $\rho > 0$ is the constant penalty parameter. The above-centralized problems can be hard to solve, so decentralized optimization methods are developed. In case the (coupling) constraints $\mathbf{c}(\mathbf{x})$ are separable, i.e., $\mathbf{c}(\mathbf{x}) = \mathbf{c}_e(\mathbf{x}) + \mathbf{c}_g(\mathbf{x}) + \mathbf{c}_h(\mathbf{x})$, and the Lagrangian does not contain the quadratic penalty term, the decentralized solution is obtained via Lagrange relaxation and optimization of three subproblems:

$$\min L_j(\mathbf{x}_j, \boldsymbol{\lambda}) = J_j(\mathbf{x}_j) + \boldsymbol{\lambda}^T \mathbf{c}_j(\mathbf{x}_j) ; j = e, g, h \quad (62)$$

In case augmented Lagrangian $L^A(\cdot)$ is applied, it cannot be written in a separable form, so decentralized optimization is done with the aid of the Alternating Direction Method of Multipliers (ADMM) [52, 57, 61]:

$$\mathbf{x}_e^{i+1} = \operatorname{argmin}_{\mathbf{x}_e} \{L^A(\mathbf{x}_e, \mathbf{x}_g^i, \mathbf{x}_h^i, \boldsymbol{\lambda}^i)\} \quad (63)$$

$$\mathbf{x}_g^{i+1} = \operatorname{argmin}_{\mathbf{x}_g} \{L^A(\mathbf{x}_e^{i+1}, \mathbf{x}_g, \mathbf{x}_h^i, \boldsymbol{\lambda}^i)\} \quad (64)$$

$$\mathbf{x}_h^{i+1} = \operatorname{argmin}_{\mathbf{x}_h} \{L^A(\mathbf{x}_e^{i+1}, \mathbf{x}_g^{i+1}, \mathbf{x}_h, \boldsymbol{\lambda}^i)\} \quad (65)$$

$$\boldsymbol{\lambda}^{i+1} = \boldsymbol{\lambda}^i + \rho \cdot \mathbf{c}(\mathbf{x}) \quad (66)$$

where i is iteration index.

On the other hand, the IES-SE model can also be represented as a sum of three objectives, where each objective is equal to the weighted sum of absolute values of differences between actual and estimated measurements. This results in a constrained Weighted Least Absolute Value (WLAV)-SE. The optimization problem is as follows [40-43, 53, 63, 64]:

$$\begin{aligned} \min J(\mathbf{x}_e, \mathbf{x}_g, \mathbf{x}_h) &= \mathbf{r}_e^T |\mathbf{z}_e - \mathbf{h}_e(\mathbf{x}_e)| + \mathbf{r}_g^T |\mathbf{z}_g - \mathbf{h}_g(\mathbf{x}_g)| + \mathbf{r}_h^T |\mathbf{z}_h - \mathbf{h}_h(\mathbf{x}_h)| \\ \text{s. t. } \mathbf{c}(\mathbf{x}) &= \mathbf{0} \end{aligned} \quad (67)$$

where $\mathbf{r}_j$ is the column vector containing diagonal elements of the corresponding measurement noise covariance matrix $\mathbf{R}_j$. The WLAV-SE formulation is known to be a robust estimator against bad data.

The modeling of gas and heat dynamic processes introduce the nonlinear state transition function $\mathbf{f}$ of gas and heat systems, which can be further combined with the IES measurement function:

$$\begin{aligned} \mathbf{x}_k &= \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{u}_k) + \mathbf{w}_k \\ \mathbf{z}_k &= \mathbf{h}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{v}_k \end{aligned} \quad (68)$$

where $\mathbf{u}$ is the input vector, $\mathbf{w}$ is the process noise which is usually assumed to be normally distributed with zero mean and covariance matrix $\mathbf{Q}$, and k denote time instant.

State transition function $\mathbf{f}(\cdot)$ can be derived from the nonlinear PDEs. To this end, numerical methods are usually used to convert the PDEs into algebraic equations. In this regard, a finite element method combined with an implicit scheme to obtain estimates in the gas system is presented in [65, 66]. In [59], Lax-Wendroff finite method is used to obtain $\mathbf{f}(\cdot)$ for the gas system, while Holt's two-parameter smoothing method is used for the electricity system. Similarly, Euler finite difference technique is used in [58] to generate algebraic equations for the gas system, and Holt's exponential smoothing technique is used to forecast electricity states. Once the algebraic equations representing the temporal system processes are developed, SE can be performed using the Kalman filter (KF) technique.

There is also a possibility of combining the Weighted Least Square (WLS) approach and KF methods. For example, WLS-SE for the electricity system is combined with an improved Extended Kalman Filter (EKF) for the gas network with transients; the two sub-problems are alternatively solved [56]. In addition, the WLS method and the KF method can be unified by exploiting the equivalence of the correction stage of the KF

method to a WLS problem. In this equivalent WLS problem, the posteriori state estimates are obtained by a compromise between the incoming real-time measurements and priori state estimates obtained from the KF prediction stage. Therefore, a WLS-form KF method can be developed to unify the SE of IES subsystems with measurement equations only (i.e., power systems and coupling units) and the SE of IES subsystems with both state transition equations and measurement equations (i.e., heat systems and natural gas systems) [49, 50].

c

An example, in which additional pseudo-measurements are not generated, is shown in *Figure 10* [62]. Assuming the electric power system (EPS) and NGS are only considered, and the SE approach is based on the multi-stage WLS method, three scenarios can be experienced (*Figure 10*). Symbols τ_e and τ_g denote, respectively, the last sampling time instant within EPS and NGS, which in turn defines the “current” time instant for SE. When $\tau_e < \tau_g$ (Scenario 1) and $\tau_e = \tau_g$ (Scenario 3), in order to model the gas dynamics, multi-snapshot measurements are used in both the NGS and EPS, linked via historical snapshots at the coupling points. However, when $\tau_e > \tau_g$ (Scenario 2), only current measurements of EPS are used because SE of EPS is based on AC load flow model with no dynamics included. In this case, only static WLS estimation of EPS states are done. Similar considerations can be applied when IES includes a heat system, resulting just in a higher number of possible scenarios.

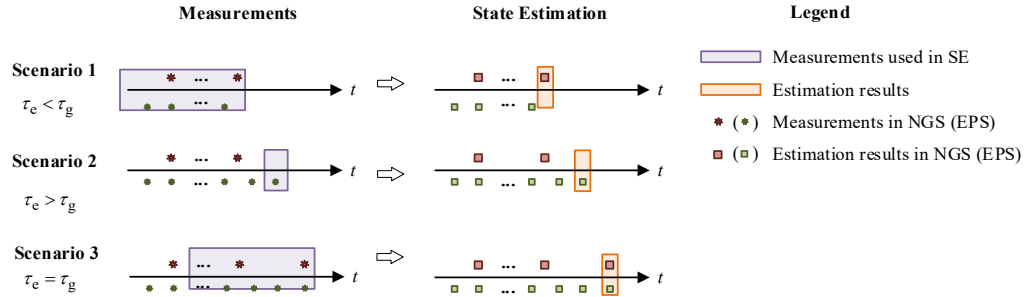


Figure 10: Illustration of the online state estimation scheme for the EPS-NGS.

4.4 Integrated Energy Systems-State Estimation Measurement Placement and Types

In this subsection, the various measurement types selected for the power, gas, and thermal systems in the IES-SE are introduced, and the measurements in multiple subsystems are discussed in detail based on the different properties of various measurement systems.

4.4.1 Measurement Types

In power systems, node voltage magnitudes, node power injections, and branch power flows are usually selected as the main measurement types. In natural gas systems, the measurement types mainly include node pressures, node injection flow rates, and pipeline flow rates. In thermal systems, pipeline mass flow rates and temperatures are the most common measurements, and almost all pipelines and nodes are equipped with these measurements. Pressure measurements are usually provided in thermal systems, but the accuracy of pressure measurement is low, and it

is greatly affected by the environment. Finally, real-time thermal power measurements are usually configured at some critical nodes, such as thermal sources or thermal exchange stations. Therefore, most thermal load nodes lack real-time thermal power measurements or only have the total thermal energy consumption of the network over a day [1].

4.4.2 Measurement Placement

With the rapid development of IES technologies, the quantity and accuracy of measurement equipment in each subsystem are being enhanced. In power systems, measurement equipment mainly consists of SCADA, Phasor Measurement Unit (PMU), and Advanced Metering Infrastructure (AMI). SCADA measurements generally include node voltage magnitudes, branch current magnitudes, branch power flows, and node power injections, have a reporting rate of 3 to 5 seconds and are present in every power system substation. PMU measurements used in SE include node voltage phasors and branch current phasors, and they report at intervals of 5 to 30 milliseconds. Extensive studies have already been done to determine locations for placing PMUs in the power system [71-74]. AMI measurements primarily come from smart meters and include nodal voltage magnitudes, branch current magnitudes, branch power flows, node power injections, and electricity consumptions. Smart meters produce outputs every 15, 30, or 60 minutes. The U.S. alone has in excess of 100 million smart meters, and their numbers are growing worldwide [75].

The measurements of the natural gas system are sampled by the turbine meters and pressure transmitters, and the collection of the measurement data can be completed instantly. The measurement equipment of thermal systems is mainly SCADA. According to the directive report of the European Union, by the end of 2016, independent thermal metering devices have been used in Europe to obtain the consumption of heating and domestic hot water in buildings such as apartments [2]. With the reform of the thermal market in China, many urban thermal systems have been equipped with more automatic calorimeters [3]. In addition, SCADA has been installed in some advanced urban thermal systems to ensure the stable operation of the thermal system.

For IES-SE, measurement placement must be carefully performed to ensure the observability of IES, a prerequisite of IES-SE. However, due to the immaturity of observability theory of IES, the measurement placement problem in the context of observability for IES has been unaddressed. Future research must be performed on the optimal placement of measurements for the observability required for IES-SE.

4.5 Integrated Energy Systems-State Estimation Observability Analysis

Observability analysis is vital to obtaining the available input measurements of SE in an IES. The observability analysis procedure must be conducted before SE is performed, and it includes the initial identification of observable states and observability restoration. The first stage searches for the maximal set of states that can be estimated with the available measurements, and the second stage provides a minimal set of additional measurements to make the system observable. Observability analysis is a necessary procedure for SE. If SE is performed without any precheck, it may encounter numerical issues. After performing observability analysis, observable states with higher accuracy can be obtained, and the convergence of SE can be ensured.

The main focus has been on observability analysis for power systems, while little work has focused on observability analysis for IES [76]. It has been confirmed that observability analysis for a multi-energy system considering the effect of the coupling units provides more reliable results than separate analysis [77]. The observability theories for SSE and DSE in power systems are different [78]. Static observability analysis paradigms consider algebraic measurement equations and result in a binary outcome, which can be classified into numerical and topological methods. Numerical methods analyze observability by checking the occurrence of zero pivots during the triangular factorization of the gain matrix [79, 80] or Gaussian elimination of the Jacobian matrix [81]. A unique solution for system states can be obtained if the gain matrix is nonsingular or if Jacobian matrix has a full column rank. The zero pivots may be misclassified due to round-off errors. Topological methods analyze observability by searching for an observable tree [82, 83] with a more complicated procedure to prevent the identification of zero pivots; they may misjudge observability in some cases [84]. The dynamic observability analysis paradigms consider the differential equations of systems and output the degree of observability (i.e., weakly, strongly, or not observable systems). The observability of a linear dynamical system can be directly determined by the rank of its observability matrix. For nonlinear dynamical systems, observability can be quantified by performing linearization at a given operating point. However, the results of this method are incorrect for highly nonlinear systems. Therefore, approaches that do not perform any linearization are proposed to assess the degree of observability. One method is to calculate the smallest singular value of the observability matrix around a given point using the Lie derivatives, which yields precise results but requires complicated computation [85, 86]. As an alternative, the degree of observability can be evaluated by computing the determinant of the empirical observability Gramian, which can reflect the observability of the nonlinear dynamics in a given domain [87, 88].

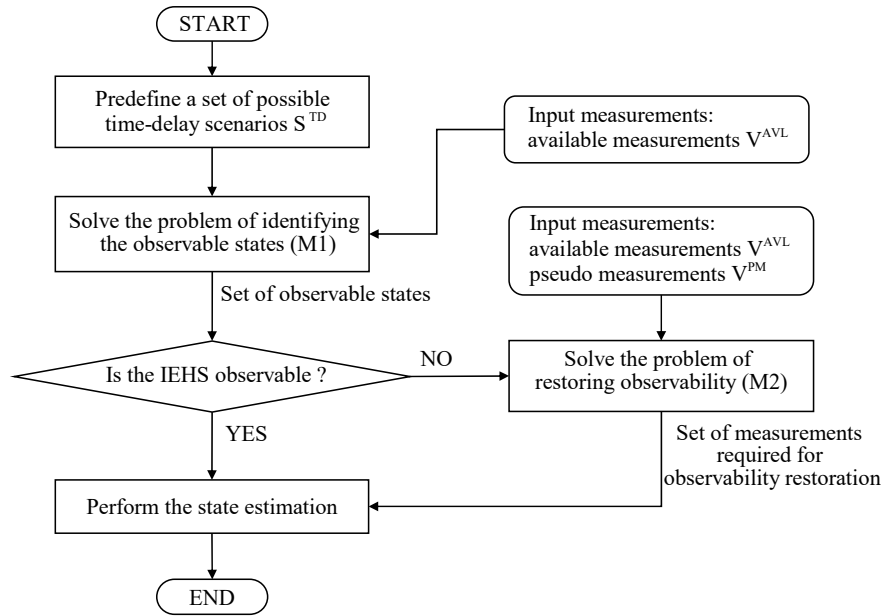


Figure 11: Procedure of observability analysis scheme for IEHS [89].

These methods presume that the measurement equations are accurately defined in advance, which is not the case for IES. Owing to the temperature dynamics in a district heating network, the set of state variables involved in the thermal dynamic equations depends on the time-delay coefficients, which are computed by the estimates of the states in turn. This implies that the measurement equations and the SE results are interdependent, making it difficult to employ conventional observability analysis methods directly to Integrated Electricity-Heat System (IEHS). An observability analysis scheme is presented in *Figure 11* for IESs which addresses the interdependency between the measurement equations and the SE results [89]. This scheme is executed to make the observability analysis results reliable for various sets of measurement equations corresponding to different time-delay scenarios so that the strong dependency of the measurement equations on the estimated results can be reduced. To this end, the problems of observable state identification and observability restoration are formulated in terms of integer linear programming, which can be handled readily by off-the-shelf solvers.

The identification of observable state variables searches for the maximal ranges of observable state variables and the corresponding relevant measurements as the input for SE. The relevant measurements are composed of real-time measurements and zero-value measurements. For an IEHS with n^{OS} observable state variables, n^{OS} measurements can be selected from the relevant measurements to form a solvable equation system. These measurements are the essential measurements. The high-level framework M1 for identification of observable state variables is given in the following form:

M1:

$$\max_{\mathbf{x}^{\text{ID}}} f^{\text{ID}}(\mathbf{x}^{\text{ID}}) \quad (69)$$

$$\text{s.t. } \mathbf{g}_1^{\text{ID}}(\mathbf{x}^{\text{ID}}; \gamma, \phi) = \mathbf{0}, \forall (\gamma, \phi) \in S^{\text{TD}} \quad (70)$$

$$\mathbf{g}_2^{\text{ID}}(\mathbf{x}^{\text{ID}}; \gamma, \phi) \leq \mathbf{0}, \forall (\gamma, \phi) \in S^{\text{TD}} \quad (71)$$

where the objective function is the number of identified observable state variables:

$$f^{\text{ID}}(\mathbf{x}^{\text{ID}}) = \sum_{i \in I} n_i \quad (72)$$

In Equations (77)–(80), f^{ID} is the objective function to be maximized; $\mathbf{x}^{\text{ID}}$ are decision variables which include the binary indicator variables for observable state variables n_i , the relevant measurements equations, the essential measurements equations, and the mapping relationships; I is the index set of states in the integrated energy system; γ and ϕ are the time-delay coefficients of heat pipes; S^{TD} is the index set of predefined time-delay scenarios; $\mathbf{g}_1^{\text{ID}}$ and $\mathbf{g}_2^{\text{ID}}$ are equality and inequality constraints imposed to make the equation system of identified relevant measurements solvable under each scenario in S^{TD} .

If the system is revealed to be unobservable by the identification step, the minimal set of irrelevant measurements and pseudo measurements should be chosen to restore the full observability of the system. For an IEHS with n^{IEHS} state variables, n^{IEHS}

essential measurements can be chosen among the relevant measurements identified by M1, and the critical measurements chosen for observability restoration. A measurement is critical if it cannot be interchanged with a nonessential measurement without affecting the solvability of the equation system. The measurements chosen for observability restoration should be critical so that they will not contaminate the SE solution. The high-level framework for observability restoration is formulated into the following form:

M2:

$$\min_{\mathbf{x}^{\text{RES}}} f^{\text{RES}}(\mathbf{x}^{\text{RES}}) \quad (73)$$

$$\text{s.t. } \mathbf{g}_1^{\text{RES}}(\mathbf{x}^{\text{RES}}; \gamma, \phi) = \mathbf{0}, \forall (\gamma, \phi) \in S^{\text{TD}} \quad (74)$$

$$\mathbf{g}_2^{\text{RES}}(\mathbf{x}^{\text{RES}}; \gamma, \phi) \leq \mathbf{0}, \forall (\gamma, \phi) \in S^{\text{TD}} \quad (75)$$

In Equations (81)–(83), f^{RES} is the objective function to be minimized, $\mathbf{x}^{\text{RES}}$ are decision variables which include the binary indicator variables for the chosen measurements, the essential measurements equations and the mapping relationships. The objective of (81) is to minimize the numbers of irrelevant measurements and pseudo measurements. The constraints $\mathbf{g}_1^{\text{RES}}$ and $\mathbf{g}_2^{\text{RES}}$ are imposed to ensure that the equation system, which is formed based on the relevant measurements identified by M1 and the measurements chosen for observability restoration, is solvable under each scenario in S^{TD} .

4.6 Integrated Energy Systems-State Estimation Methods

In this subsection, a thorough, yet not comprehensive, review of the IES-SE algorithms is presented. These algorithms are classified and discussed based on various criteria, such as problem formulations and computing architectures. The IES-SE considering different uncertainties and sampling resolutions are also presented.

4.6.1 Based on Problem Formulation

IES can be mathematically characterized by a set of both algebraic and differential equations, whose discrete-time representation is written as

$$\begin{aligned} \mathbf{x}_k &= \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{u}_k) + \mathbf{w}_k \\ \mathbf{z}_k &= \mathbf{h}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{v}_k \end{aligned} \quad (76)$$

where the subscript k denotes the k th time instant, $\mathbf{f}(\cdot)$ and $\mathbf{h}(\cdot)$ are the nonlinear state transition and measurement functions, respectively. $\mathbf{x}_k$ is the state vector. $\mathbf{u}_k$ is the input vector, $\mathbf{z}_k$ is the measurement vector, and $\mathbf{w}_k$ and $\mathbf{v}_k$ are the noises for the state transition process and measurement, which are usually assumed to be normally distributed with zero mean and covariance matrices, $\mathbf{Q}_k$ and $\mathbf{R}_k$, respectively.

Based on the modeling and estimation approach used in IES-SE, the existing SE algorithms can be classified into three categories:

- Static State Estimation
- Quasi-dynamic State Estimation

- Dynamic State Estimation
- Static-Dynamic Hybrid State Estimation

The detailed explanation of each of these categories is as follows:

- a) **Static State Estimation Algorithm:** SSE algorithms are carried out under the quasi-steady state operating condition, where the state transition information is fully neglected and only the nonlinear measurement functions are considered. In this case, SSE has no memory of historical states and no capability to track the system dynamics, only the latest set of available measurements that can solely make the state vector observable is required to perform an SSE.

The equality constrained WLS method is a commonly used approach to tackle the SE problem for IESs [51, 57, 60, 90-93] with the interaction of natural gas, heat, and power systems. To counter the possible measurement outliers, WLAV estimator is adopted to take advantage of its better robustness [53, 64].

To deal with the nonlinearity of IES-SE, several bilinear SE methods are proposed in [40-43, 52, 94-96] by introducing auxiliary variables. The auxiliary variables are estimated using a linear WLAV estimator with the available measurements, after a nonlinear transformation of which the state estimates can be solved by a linear WLS or WLAV estimator.

- b) **Quasi-Dynamic State Estimation Algorithm:** Since the time constants of both heat and natural gas systems are much larger than that of the power system, these systems have much slower dynamic characteristics. The heat and gas pipeline dynamics couple the current nodal states with the historical ones. In this condition, the SSE model, which treats the heat and gas transfer as quasi-steady processes, will lead to poor estimation accuracy. To improve the estimation quality, the dynamic models of the heat and natural gas systems should be considered in IES SE.

Several researchers have extended the existing SSE methods to integrate the thermal and gas dynamics, yielding Quasi-Dynamic State Estimation [97]. Based on the dynamic models of the pipelines, the progress of the fluids of gas and water in the corresponding gas and heat systems can be tracked as the flows vary, thus taking the historical system states into account. These gas and heat dynamic constraints of each pipeline are incorporated together with the steady-state constraints in an equality-constrained static estimator, where the available measurements can be utilized to track the states of IESs [22, 54, 98].

- c) **Dynamic State Estimation Algorithms:** To better track the dynamics of IESs, recursive estimators, which are inherently capable of propagating the state statistics, are investigated to perform IES SE and are referred to as IES-DSE.

The nonlinear filters within KF framework, such as EKF [59], cubature KF [99], and unscented KF [100], allow propagating states at the previous time steps based on the dynamic models of the pipelines for state prediction. Subsequently, these predictions are utilized together with the measurement information to update the state estimates and the covariance matrix.

- d) **Static-Dynamic Hybrid State Estimation Algorithms:** In IES, it is common to encounter situations where certain subsystems are described by pure algebraic equations, i.e., measurement equations only (e.g., electric power systems, coupling units, etc.), and certain subsystems are described by differential-algebraic equations, i.e., state transition equations and measurement equations. As two types of subsystems are suitable for different types of SE techniques, there is a need to either coordinate or to fuse different SE techniques. For decentralized IES-SE, the natural approach is to perform different types of SEs for different subsystems, and exchange information between them to yield solutions with consensus [43]. For centralized ISE-SE, however, an integrated estimation algorithm must be developed for the whole system. This can be achieved by exploiting the equivalence between the correction stage of the KF approach for DSE and the WLS approach for SSE. The correction stage of KF can be viewed as a WLS problem where the a priori state estimates from the prediction stage act as “pseudo-measurements” to be fused with the incoming actual measurements, from which the a posteriori state estimates are obtained. In [49, 50], a WLS-form iterated EKF is developed to fuse measurements from subsystems with state transition equations with measurements from subsystems without state transition equations. It is a centralized hybrid algorithm applicable to IES with subsystems of different natures.

4.6.2 Based on Computing Architecture

In the aspect of IES-SE computation architectures, the concerned research can be divided into two types, i.e., centralized and decentralized.

- a) **Centralized SE Algorithms:** For centralized IES-SE, [51] first introduces the conventional WLS estimator in the power system to the integrated heat and power system, where the state variables and measurements in different energy systems are integrated to solve a unified SE problem. Similar to the work presented in [22, 40, 41, 51, 53, 60, 90, 92-95, 100-102], IES-SE is performed in a centralized way by estimating the state variables in different energy systems simultaneously. These studies point out that the centralized SE can significantly improve the estimation accuracy compared with the individual ones since the measurement redundancy is improved by the pseudo measurements of coupling components.
- b) **Decentralized SE Algorithms:** Although achieving good accuracy, centralized SE faces many challenges, such as privacy issues [61], different sampling resolutions [22, 99], and time-consuming problems [5]. To overcome these challenges, the decentralized framework is preferred, where SE of different energy systems are performed individually and the boundary information of coupling components is exchanged to ensure a global convergence [42, 43, 52, 57, 59, 61, 64, 96, 103].

ADMM is a widely used method to handle the decentralized SE problem [52, 103] where the estimation is performed in a serial way, i.e., only one energy system is estimated at a time and the boundary information of components is updated for the next system to be estimated. This serial way can share the boundary information timely and thus achieve fast convergence. However, it

may suffer from a waste of computation resources. To address the issue, [57] proposes a synchronous parallel way to perform the estimation of each energy system at the same time, and the boundary information is exchanged when all local estimations are completed. Nevertheless, the synchronous parallel way may suffer from an unbalance in local estimation time. To reduce idle time, an asynchronous parallel way is adopted to perform decentralized SE [15, 48].

4.6.3 Consideration of Uncertainties

In practice, IESs are exposed to various unavoidable uncertainties which may deteriorate the estimation quality. The uncertainties, such as those of measurements, communication, and system model, should be fully considered in IES SE to ensure the accuracy of estimation results.

The uncertainties associated with measurement information are the first concern of IES-SE since its accuracy depends greatly on the precision of measurements. Unlike the deterministic SE in most of the existing researches, the interval SE [17] provides accurate bounds of state variables by using uncertain measurement information, especially pseudo-measurements, modeled as particular intervals. This facilitates the indication of whether IES is operated securely or not. The performance deteriorates when measurement outliers exist. To avoid this, it is crucial to implement bad data detection and identification; calling for robust estimators to suppress the adverse impact of bad data. For estimators without inherent robustness against bad data, an additional bad data processing should be embedded to detect and identify gross errors. For instance, the largest normalized residual test is employed following a WLS estimation in [1, 19] to remove the suspected bad data. However, it cannot effectively identify the multiple gross errors. To deal with this, other alternative estimators endowed with robustness to measurement outliers have been proposed, such as WLAV estimators [14, 28]. The uncertainty of communication also has a great impact on the estimation accuracy of IES.

Poor communication quality will not only directly lead to low measurement precision, but also result in inaccurate data interaction in decentralized IES-SE. The impact of communication packet loss on estimation accuracy is investigated in [45]. However, there still exists a large research gap in IES-SE considering communication uncertainties.

Another practical issue contributing to ISE-SE errors is the uncertainty of system models. In IES-SE's performance of filtering measurement noise and bad data highly depends on the accuracy of system models. In power systems, it is well known that electric network parameters are never fully accurate. A large volume of research has been dedicated to the detection, identification, and correction of modeling errors in power systems such as [104-106]. In IES, the model parameters of heat networks, gas networks, and coupling units are also prone to inaccuracies and errors. They can bias SE results and even lead to the misidentification of good measurements as bad data. In-depth investigation is required to handle the impact of modeling uncertainty of ISE-SE.

4.6.4 Consideration of Different Sampling Resolutions

Due to the different applications, the sampling resolutions of measurements in diverse energy systems are inconsistent with each other. In the case of a power system, since

its fast dynamics may cause severe instability issues, data are sampled with high frequency and transmitted with a dedicated wired communication network (e.g., optical network). For instance, the sampling rate of the SCADA system usually ranges from a few seconds to a minute. In contrast, for natural gas systems and heat systems, the measurement information is collected for billing purposes in most circumstances, where lower sampling rates are acceptable. In this case, the sampling resolutions of these systems are usually minutes to hours.

To deal with the multiple sampling resolution issue, the following two types of methods are exploited:

- a) **SE with Sampling Interpolation:** The sampling interpolation approaches such as linear extrapolation and linear interpolation aim at generating intermediate pseudo-measurements to make the sampling rates of the different systems consistent [14]. In this way, both centralized and decentralized IES-SEs can be executed under a unified frequency.
- b) **Multi-time-scale SE:** This type of approach allows the implementation of SE in different energy systems at different frequencies [11, 15]. The simultaneous SEs of all energy systems, whether centralized or decentralized, are carried out only if their measurements are received at the same time. Otherwise, the individual SE for the power system will be performed by using the last state estimates of the natural gas and heat systems or their approximation obtained based on the dynamic models.

4.6.5 Data Exchange Between Individual Systems

Required data exchange for SE of IES is driven by the connectivity of individual energy subsystems, applied SE solution methodology, observability of individual subsystems, applied protocol for data exchange, and communication security level. To optimize the communication burden, it is necessary to recognize which data are essential and have to be exchanged, i.e., there is no redundant, unnecessary information. In the case of a centralized SE, modelling data of all networks together with measurement locations, types and values need to be collected in a single control center. The volume of data exchange is high and communication channels need to have sufficient capacity and speed particularly when the resolution rates are high. This makes centralized SE of IES practically difficult to implement. On the other hand, in case of decentralized SE, data exchange between centers is smaller and can depend on the solution methodology. For example, decentralized SE based on primal-dual decomposition in [26] and/or ADMM methods [52, 57, 61] require transfer of estimated subsystem states at the current time instant.

At the moment, different energy subsystems that compose IES are operated by different energy system operators, leading to communication inconvenience and privacy protection among subsystems. Unreliability of communication between control centers represents an obstacle for decentralized SE of IES. To facilitate communication between the control centers of electricity, gas and heat sectors, the standardization of communication protocols across different energy sectors is necessary. For instance, Inter-Control Center Communications Protocol (ICCP) or IEC 60870/TASE.2) is considered as the standard worldwide for control-center communication in the electric power sector, allowing the exchange of real-time and

historical power system monitoring and control data [107, 108]. For future integrated energy systems, this must be aligned with communication protocols used in other energy sectors. Developing a comprehensive, fully compliant standard for real-time data exchange between control centers of different energy sectors could be considered preferable. It is also important to secure communication from cyber-attacks that can exert disruptions and damage ranging from service interruptions, blackouts, economic losses, and personnel life threats. Secure communications enable exchange of trustworthy information between control centers.

Non-observability of a subsystem can lead to the SE failure of the entire IES, which has to be avoided. In such a case, subsystem states cannot be estimated and there is no scope for data transfer; the only remaining solution is independent SE of the observable subsystems.

4.7 Difference between Integrated Energy Systems-State Estimation Algorithms

In this section, differences between IES-SE and PS-SE are introduced, which are respectively reflected in the following three aspects:

4.7.1 Modeling Differences

SE problem is a data filtering problem based on the physical model and raw measurements. Compared to the maturity of traditional PS-SE modeling theory, although the research of thermal and natural gas system SE is blooming, SE models applied in thermal and natural gas systems are often oversimplified, limiting practical applications, and thus IES-SE requires further improvement.

The static model of power systems is often based on Kirchhoff's voltage and current law. The main components of electrical networks such as lines and transformers are described by the general branch model [103]. Generators and loads are described by the grounding branch model, developing a network analysis method that takes node voltages, node power injections, and branch power flows as variables [47, 103]. As the transient process of power systems is very short, the static model is sufficient to meet the requirement of SE accuracy [103]. However, due to the large transient time constant of the thermal network and gas network, it needs a long transient process to restore the balance under an external disturbance. If only the static model is used, it will result in large estimation errors, and thus it is essential to choose a suitable system model according to different operating conditions and simulation requirements.

At present, static models [57, 94] are mainly used in the research of electricity-heat SE. Research on electricity-heat DSE needs further investigation, which is carried out only when the heat transfer delay [103] is concerned in the pipeline thermal model. In the research of electricity-gas SE, there are three types of gas network SE models:

- a) The static model of the gas network [41, 93] whose main characteristic is to ignore the energy loss of pipeline friction and only consider the relationship between the pressure at both ends of the pipeline and pipeline flow rate.
- b) The transient model of the gas network, in which PDE is solved by the implicit difference method, during which different step sizes are set. The calculation burden of this method increases significantly with the increase in the number of nodes.
- c) The gas network model is based on energy circuit theory, where the phasor method is introduced to map the circuit from the time domain to the frequency

domain. The physical equations can be expressed in a matrix form and solved more uniformly.

4.7.2 Measurement Differences

The measurement differences between IES-SE and PS-SE are as follows:

- a) In addition to the traditional power system measurements, the pressure and flow rate measurements in natural gas systems and the pressure, flow rate, and temperature measurements in the thermal systems are required.
- b) Due to the influence of the operating environment of sensors and the limited instrument automation [103], the measurement errors and the probability of obtaining bad data in gas and thermal systems are usually higher than those in power systems.
- c) Compared to electrical networks, the transient processes of heat and gas networks are longer because of the large inertia of systems, and the measurement data is time-delayed, so instantaneous mutation usually does not happen.
- d) The sampling frequency of the measurement data (updated in seconds or minutes) is relatively high in the power system because the fast dynamic characteristics of the power system may lead to the loss of stability. The measurement data of the gas and heat networks are still used for billing purposes [96] in most cases, and its sampling frequency is relatively lower than that in the electrical network.
- e) The coupling measurements in IES can be used to measure both sides of the interconnected systems, thus reducing the number and cost of measuring equipment and improving the observability of the system. In addition, when some of the key measurements are missing, the overall IES may become unobservable [103].

5. Demonstration Project in China

IEMS is the brain of IESs to manage and control the multi-energy flow, which can really break down the barriers between different energy systems and coordinate them to achieve the optimal economic, secure, low-carbon operation of IESs. China leads the global study and applications of IEMS. The team from Tsinghua University designed and developed the world's first IEMS and it applied in many pilot demonstration projects covering park-level and city-level IESs [35-38, 109, 110].

Online SE is the basic function of the IEMS, which can provide real-time, reliable, consistent, and complete network state information for the optimal operation and security analysis of the multi-energy network. For the practical implementation, there are three main challenges:

- a) Various multi-energy flow system scenarios present significant differences in the energy flow types, coupling units, and network scales. The developed SE function needs to meet the needs of the various multi-energy flow systems and achieve flexible adaptability.
- b) Different energy systems have different time scales and time resolutions, which are also related to the operation scenario. Therefore, it is necessary to consider a multi-time scale integrated SE model suitable for different applications, adapting to both steady-state and dynamic operating conditions.

- c) It is difficult to ensure the observability of a weakly-measured multi-energy flow network. In practice, the measurement configurations of heating networks, cooling networks, and gas networks are weaker than electric power systems, which may lead to a lack of system observability. Therefore, it is necessary to propose methods to supplement the measurement to meet the requirements of observability.

To realize the online SE, a multi-scene adaptive online SE functional framework of the multi-energy network is designed in [32] as shown in *Figure 12*. The developed SE functional framework was applied in an industrial park in Beijing, China [69].

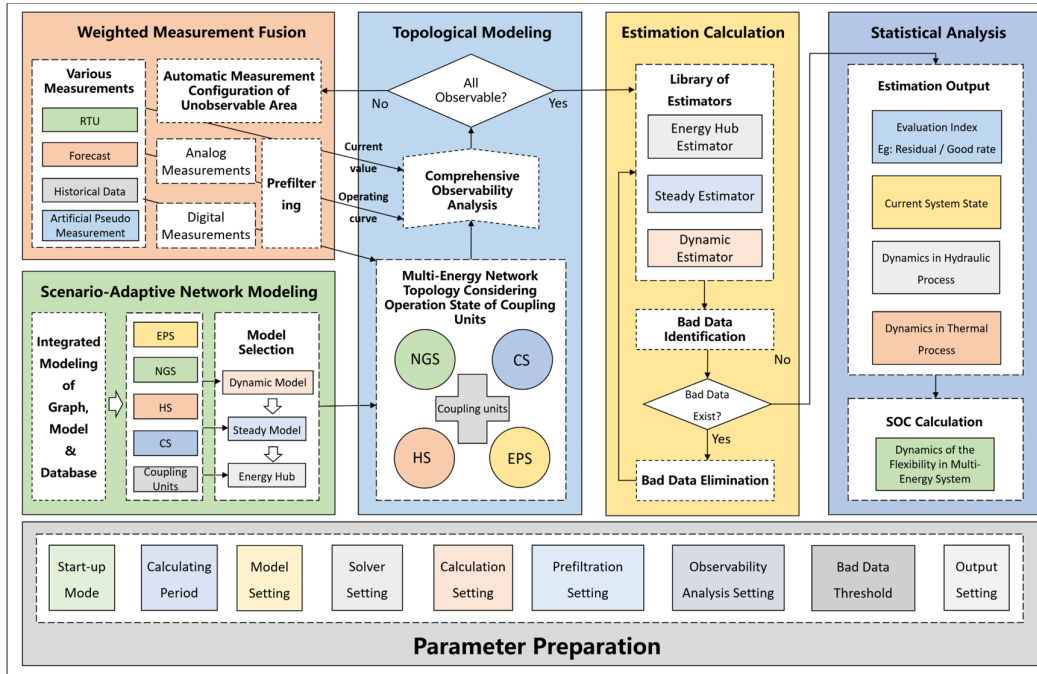


Figure 12: Flow chart for SE function of the integrated energy management system.

5.1 System Description

The SE framework consists of the following five modules:

- Weighted Measurement Fusion:** This module supports reading various categories of measurements from IEMS. In addition to real-time measurements from SCADA, other measurements including the forecast data, historical data, and artificial pseudo measurements are considered in this module to solve the problem of low redundancy of measurement configuration in multi-energy networks. The weight of each kind of measurement is set by the operator.
- Scenario-adaptive Network Modeling:** This module generates different kinds of energy network models adapting to various scenarios with different time scales and operating conditions.
- Topological Modeling:** Based on the measurements and network model, this module generates the topological model for SE calculation which includes:
 - Comprehensive topology analysis** is the basis of SE calculation. The topology of the power grid, heating network, cooling network, and gas

network is analyzed, and the integrated energy network topology is generated according to the operation state of coupling units.

- **Comprehensive observability analysis** identifies the maximum observable area of the whole multi-energy system and adds new pseudo measurement to the unobservable area until the observability of the whole system is achieved.
- d) **State Estimation:** This module carries out SE computations, including the bad data identification. To adapt to different scenarios of the multi-energy flow system, the integrated SE function library includes an energy hub estimator, a steady-state estimator, and a dynamic state estimator. Based on the scenario-adaptive network model, real-time, reliable, consistent, and complete state information results of the multi-energy network are obtained by corresponding estimators. However, bad data may exist in the measurement data and cannot be eliminated by prefilter identification. The bad data will reduce the accuracy of SE results, so bad data identification is also performed during the estimation process.
- e) **Statistical Analysis:** After the calculation of SE, the module generates the statistical indicators including estimation residual errors and qualified rates, and outputs real-time, reliable, consistent, and complete operating state information of the multi-energy system, including steady-state data and dynamic state data. Besides, it also supports the real-time evaluation of the energy storage state of the pipe network.

5.2 Test Results in an Industrial Park

The developed SE function has been applied in a multi-energy system inside an industrial park in Beijing, China, since 2019, when the world first IEMS was developed and applied. The application successfully verifies the effect of the steady-state IES-SE. *Figure 13* shows the topology of the energy system of the park, which is composed of electricity, heating, cooling, and natural gas systems. The supply network and return network of both the heating and cooling systems are symmetrical. The energy supply devices in this park include a gas turbine, lithium bromide refrigeration, gas boiler, direct-fired unit, electric refrigerators, photovoltaic, wind power, electric energy storage, and other components, among which the cooling sources, heating sources, and some electrical sources are in the energy station, providing energy for the five buildings (Load1~Load5) in the park.

- a) **Typical Estimation Results in Summer:** In summer, the natural gas network and heat supply network are not put into operation, so the energy system of the park is a combined cold and power network. Two electric refrigerators supply the cooling water to two buildings (Load 4 and Load 5). The supply flow of the two electric refrigerators converges in the energy station, then injects into a water distributor from a water supply pipe and sends to the two loads respectively. The return flow of the two loads converges through the water collector, then is sent to the return node of the two electric refrigerators through a return pipe. Due to the pipeline connecting the water collector and water distributor with the load (the blue dotted line in *Figure 13*) being too short, they are neglected in actual operation, and only the supply pipeline and

return pipeline are considered (the blue line in *Figure 13*). The measurement configuration includes the mass flow rate, supply and returns water temperature at the ends of the cold source and cooling load, and the head of the pipeline.

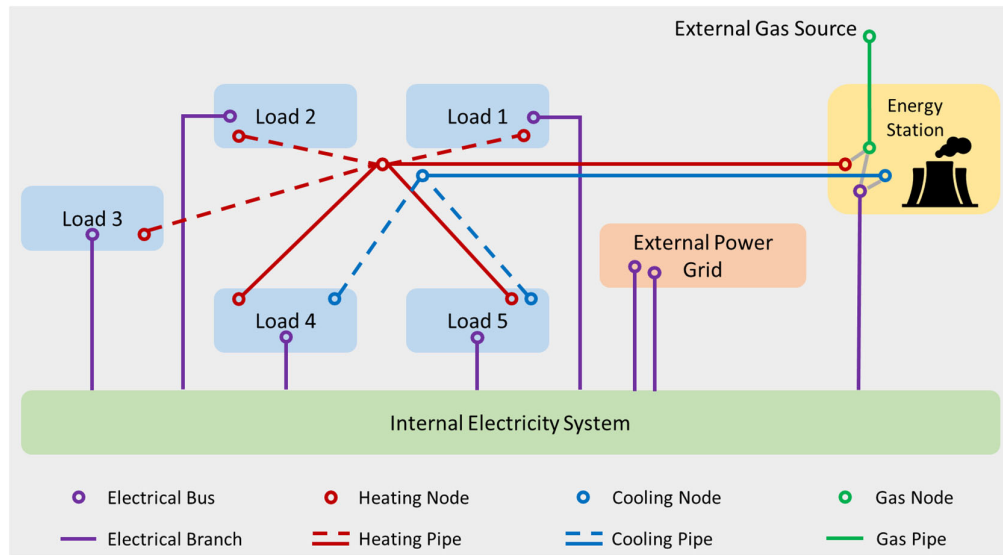


Figure 13: Structure of the multi-energy system inside an industrial park in Beijing, China.

Consider 15:55 on July 22, 2019, as an example. The SE and bad data identification results of typical sections are given in *Table 4* and *Table 5*. The residual sum of squares (RSS) of good measurements, bad measurements and all measurements are presented.

There are seven identified bad measurements in the multi-energy system. Four of them are in the cooling network, including one missing temperature of which the measurement value is 0, one wrong flow direction, and two large measurement errors which contribute the major part in the RSS of all measurements. SE function successfully identifies the bad data, eliminates them, and obtains the correct estimation results.

Table 4: Estimate results of the operation section at 15:55 on July 22, 2019

Statics	Electricity System	Cooling System	Multi-energy System
Residual sum of squares of good measurements	8.43	31.16	39.59
Number of the bad measurements	3	4	7
Residual sum of squares of bad measurements	0.94	466284.62	466285.56
Residual sum of squares of all measurements	9.37	466315.78	466325.16

Table 5: Bad data identification result of the cooling network

Variable	Measurement	Estimation
The inlet temperature of the water supply pipe from the electric refrigerator to the water distributor /°C	0	10.0
The mass flow rate of the water supply pipe from the electric refrigerator to the water distributor /(kg/s)	-338.94	343.80
The mass flow rate of the water return pipe from the water collector to the electric refrigerator/(kg/s)	350.25	343.80
The inlet temperature of the cooling load in Load 5 /°C	7	10.0

- b) **Typical Estimation Results in Winter:** In winter, the energy station provides heating services, to five buildings (Load1~Load5) with a gas turbine, so the multi-energy system is a combined power and heat network. The measurement configuration of the supply pipeline, return pipeline and the energy station is same as the cooling network in summer. As for the loads, only Load 4 and Load 5 are equipped with meters to measure the temperature and mass flow rate of supply and return water. The states of Load 1, Load 2, Load 3 and pipelines connecting to them cannot be estimated due to the lack of measurements. Thus, only the operation of Load 4, Load 5, energy station and the pipelines connecting to them (the red line in *Figure 13*) are considered in the SE, while other parts of the system are neglected. However, the mass flow rate and return temperature of Load 1, Load 2 and Load 3 are given as pseudo measurements to support the calculation of the SE, of which the value is given according to the historical operation results. The weights of those pseudo measurements are set as one-tenth of the weight of the same type of variable to decrease their influence of the SE results. Also, the choice of weight is given by the operator based on practical experience with the system. The heating network pipeline of the park is short, and the time delay is not obvious, so the steady-state estimation is selected. In the actual operation, SE is carried out every 1 minute which is the same as the field sampling rate of SCADA.
- c) *Figure 14* shows the residual estimation results of 24 hours starting from 16:00 on January 18, 2020. The developed SE function runs well continuously and effectively eliminates bad data. The bad measurements deviate greatly from the real value, while the good measurements deviate slightly from the real value. Though the number of the bad measurements is much smaller than the good measurements, since the SE obtains the correct estimation results close to the real value, the RSS of the bad measurements is much higher than that of the good measurements in most cases.

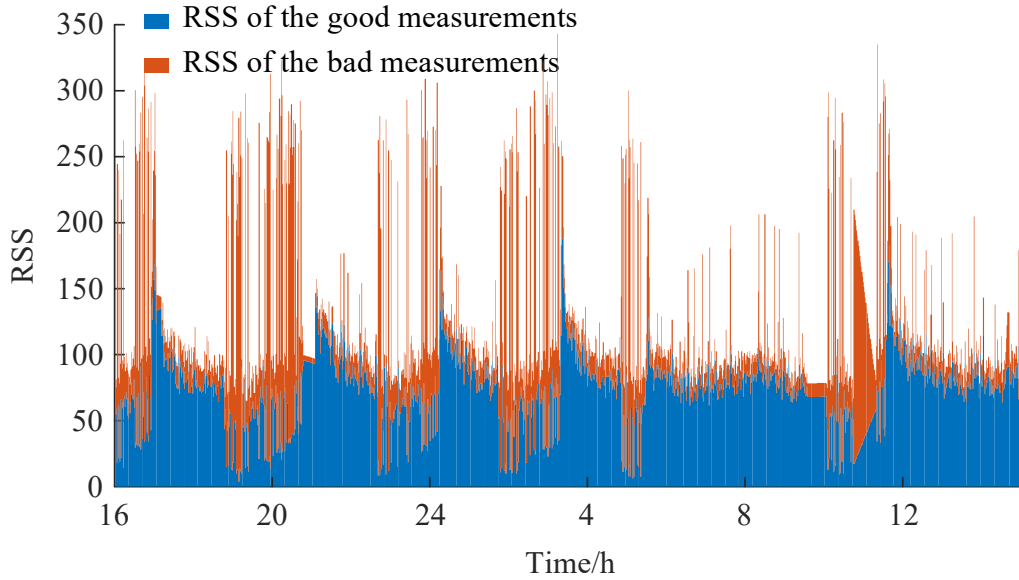


Figure 14: Residual sum of squares in a typical day of operation.

5.3 Test Results in a City-Scale IES

In a large-scale district heating network or natural gas network, the dynamics have to be considered. Take the large-scale district heating system as an example, different from the industrial parks, the supply temperature of the heat source adjusts to meet the varying load, while pipes are too long to ignore the time delay [35, 36].

DSE is demonstrated in a typical demonstration project in Changchun City, Jilin Province, which involves a city-scale IES. In this demonstration area, 8 CHP units with an installed capacity of 1,820 MW, 8 wind farms with an installed capacity of 1,288 MW, and 2 electric-heating substations with an installed capacity of 46 MW are integrated to supply power and heat. The heat is delivered by a heating network that has 400 heat exchange stations and pipelines over 268 km (containing water of roughly 300 thousand tons) to a load of 20 million m². The power and heat system involved in this demonstration project contains a 621-bus-817-branch electricity network and a 376-node-508-branch heating network [20].

Specifically, DSE is executed every 5 minutes, and each execution is finished in 30 seconds (typical). A period lasting for 8 hours is considered. The average residual sum of squares of all measurements at the period is 0.4434 and is smaller than 1, which means DSE successfully tracks the temperature dynamics in the DHS.

The real-time HSOC of the heating network is also obtained, as shown in Figure 15. The HSOC increases slowly and reaches a peak value at approximately 3.75 h and then decreases sharply. Figure 15 shows the total charging power $\psi_h(t)$ is positive when the HSOC charges and becomes negative when the HSOC discharges. It can be also found that when the absolute value of the total charging power is higher, the HSOC changes faster.

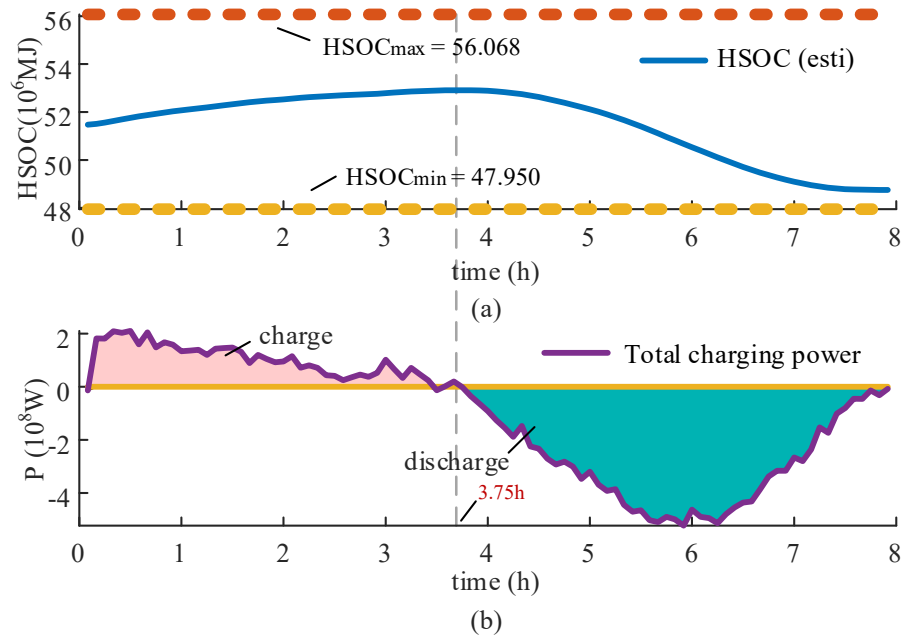


Figure 15: (a) Estimation result of the total HSOC in the DHS; (b) charging and discharging process of the DHS.

6. Future Works and Open Challenges

This section presents the ideas for future works and a few open challenges related to IES-SE:

- Measurement Placement and Communication Design in IES-SE:** The availability of measurement data is a key to the proper implementation of IES-SE. This relies on a measurement and communication infrastructure that collects and transmits measurement data. As the current heat systems and natural gas systems lack the proper monitoring infrastructure for system observability, it can be foreseen that investment will be made in this domain to allow advanced functionalities of IES. The optimal design of measurement and communication configurations that delivers observability under normal and faulty conditions while minimizing system upgrade cost and end-to-end data transfer delay is an open area calling for systematic research. Due to the multi-time scale complexity of the IES, the existing observability theory may not work and thus new theory and approach are needed.
- IES Model Improvement:** one major challenge for the IES SE is to correctly represent the couplings among different subsystems. These involve multi-time scale coupling relationships via either differential algebraic equations or algebraic equations only or the combinations of them. There is a need to develop proper criteria or reduced order models so that the important states are captured.
- IES Model Uncertainties:** While most of the existing methods of IES-SE consider the uncertainty of measurement data, it is assumed (by default) that the IES-model is accurate. However, in practice, the model parameters or even topologies are subjected to errors, which in turn deteriorates the performance of IES-SE. Furthermore, when there is an inconsistency between the model and the

measurement data, it is challenging to determine which is the true error source. Therefore, the model identification problem must be considered coordinately with IES-SE.

- **Modeling of Transportation Systems and Electric Vehicles:** One of the key findings of this report is that the current research effort of IES-SE exclusively focuses on the interaction between electric power systems, heating and cooling systems, and natural gas systems, while the role of transportation systems has largely been missed. With the growing electrification of the transportation sector, there is great potential for its coordination with other energy systems, especially the power systems, to enhance energy efficiency and reliability. In this context, the transportation systems, including large electric vehicle fleets, should be taken into the scope of IES-SE. However, the physics of transportation systems is less clear than and more distinct from the other energy systems, requiring major innovation in IES-SE for their incorporation.

7. References

- [1] M. Riipinen, "District heating and cooling in Helsinki," in *International Energy Agency CHP/DHC Collaborative and Clean Energy. Paper presented at the Ministerial CHP/DHC Working Group Joint Workshop*, 2013.
- [2] C. Kang *et al.*, "Balance of power: toward a more environmentally friendly, efficient, and effective integration of energy systems in China," *IEEE Power and Energy Magazine*, vol. 11, no. 5, pp. 56-64, 2013.
- [3] A. J. Conejo, S. Chen, and G. E. Constante, "Operations and long-term expansion planning of natural-gas and power systems: A market perspective," *Proceedings of the IEEE*, vol. 108, no. 9, pp. 1541-1557, 2020.
- [4] N. Tanaka, "World energy outlook 2010," *International Energy Agency. Paris: IEA*, 2010.
- [5] H. Jin, H. Hong, B. Wang, W. Han, and R. Lin, "A new principle of synthetic cascade utilization of chemical energy and physical energy," *Science in China Series E: Technological Sciences*, vol. 48, pp. 163-179, 2005.
- [6] H. Sun, Z. Pan, and Q. Guo, "Energy management for multi-energy flow: challenges and prospects," *Automation of Electric Power Systems*, vol. 40, no. 15, pp. 1-8, 2016.
- [7] P. Stanislav, K. Bryan, and M. Tihomir, "Smart grids better with integrated energy system," in *IEEE Electrical Power & Energy Conference (EPEC)*, 2009, pp. 1-8.
- [8] P. Mancarella, "MES (multi-energy systems): An overview of concepts and evaluation models," *Energy*, vol. 65, pp. 1-17, 2014.
- [9] M. O'Malley *et al.*, "Energy systems integration. Defining and describing the value proposition," National Renewable Energy Lab.(NREL), Golden, CO (United States), 2016.
- [10] J. Wu, J. Yan, H. Jia, N. Hatzigargyriou, N. Djilali, and H. Sun, "Integrated energy systems," *Applied Energy*, vol. 167, pp. 155-157, 2016.
- [11] Y. Wang, B. Zeng, J. Guo, J. Shi, and J. Zhang, "Multi-energy flow calculation method for integrated energy system containing electricity, heat and gas," *Power System Technology*, vol. 40, no. 10, pp. 2942-2950, 2016.
- [12] Q. Qiu and X. Luo, *Circuits (5th Edition)*. Beijing: Higher Education Press, 2011.

- [13] Y. He and Z. Wen, *Power System Analysis (Vol. 2) (Fourth Edition)*. Wuhan: Huazhong University of Science and Technology Press, 2016.
- [14] X. Liu, J. Wu, N. Jenkins, and A. Bagdanavicius, "Combined analysis of electricity and heat networks," *Applied Energy*, vol. 162, pp. 1238-1250, 2016.
- [15] H. Zhao, "Analysis, modelling and operational optimization of district heating systems," Report Number: NEI-DK-2023, Denmark, 1995.
- [16] M. Jiang, S. Wang, and Z. Zeng, "Simulation and analysis of gas transmission and distribution network," ed: Beijing: Petroleum Industry Press, 1995.
- [17] D. De Wolf and Y. Smeers, "The gas transmission problem solved by an extension of the simplex algorithm," *Management Science*, vol. 46, no. 11, pp. 1454-1465, 2000.
- [18] A. Abur and A. G. Exposito, *Power system state estimation: theory and implementation*. CRC press, 2004.
- [19] B. E. Larock, R. W. Jeppson, and G. Z. Watters, *Hydraulics of pipeline systems*. CRC press, 1999.
- [20] B. Chen *et al.*, "Energy-Circuit-Based Integrated Energy Management System: Theory, Implementation, and Application," *Proceedings of the IEEE*, vol. 110, no. 12, pp. 1897-1926, 2022.
- [21] A. Osiadacz, "Simulation and analysis of gas networks," *Gulf Pub Co*, ISBN-13:978-0872018440 1987.
- [22] T. Sheng, G. Yin, Q. Guo, H. Sun, and Z. Pan, "A hybrid state estimation approach for integrated heat and electricity networks considering time-scale characteristics," *Journal of Modern Power Systems and Clean Energy*, vol. 8, no. 4, pp. 636-645, 2020.
- [23] T. Sheng *et al.*, "State estimation approach for combined heat and electric networks," *CSEE Journal of Power and Energy Systems*, vol. 8, no. 1, pp. 225-237, 2020.
- [24] X. Liu, "Combined Analysis of Electricity and Heat Networks," Ph.D. Thesis, Institute of Energy, Cardiff University, 2013.
- [25] P. M. Coelho and C. Pinho, "Considerations about equations for steady state flow in natural gas pipelines," *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, vol. 29, no. 3, pp. 262-273, 2007.
- [26] Y. Feng, V. Levi, D. Cetenovic, J. Zhao, P. Taylor, and V. Terzija, "Primal-dual decomposed state estimation for multi-energy systems leveraging variational Bayesian approximation," *Early Access, IEEE Transactions on Smart Grid*, 2023.
- [27] P. Jany and G. Thieleke, *Thermodynamik für Ingenieure: ein Lehr-und Arbeitsbuch für das Studium*. Springer-Verlag, 2011.
- [28] I. B. Hassine and U. Eicker, "Impact of load structure variation and solar thermal energy integration on an existing district heating network," *Applied Thermal Engineering*, vol. 50, no. 2, pp. 1437-1446, 2013.
- [29] BSI Standards Publication, "Natural gas-Determination of Composition and Associated Uncertainty by Gas Chromatography: Part 1-General Guidelines and Calculation of Composition," *Standards Policy and Strategy Committee*, November 2012.
- [30] J. S. Stecki and D. Davis, "Fluid transmission lines—distributed parameter models Part 1: A review of the state of the art," *Proc. Institution of Mechanical*

- Engineers, Part A: Power and Process Engineering*, vol. 200, no. 4, pp. 215-228, 1986.
- [31] B. C. Dorin and D. Toma-Leonida, "On modelling and simulating natural gas transmission systems (Part I)," *Journal of Control Engineering and Applied Informatics*, vol. 10, no. 3, pp. 27-36, 2008.
- [32] C. M. Correa-Posada and P. Sanchez-Martin, "Integrated power and natural gas model for energy adequacy in short-term operation," *IEEE Transactions on Power Systems*, vol. 30, no. 6, pp. 3347-3355, 2014.
- [33] "Numerical methods for partial differential equations (2022) Wikipedia." https://en.wikipedia.org/wiki/Numerical_methods_for_partial_differential_equations (accessed August 25, 2023).
- [34] A. Tomasgard, F. Rømo, M. Fodstad, and K. Midthun, "Optimization models for the natural gas value chain," *Geometric modelling, numerical simulation, and optimization: Applied mathematics at SINTEF*, pp. 521-558, 2007.
- [35] Z. Li, W. Wu, M. Shahidehpour, J. Wang, and B. Zhang, "Combined heat and power dispatch considering pipeline energy storage of district heating network," *IEEE Transactions on Sustainable Energy*, vol. 7, no. 1, pp. 12-22, 2015.
- [36] Z. Li, W. Wu, J. Wang, B. Zhang, and T. Zheng, "Transmission-constrained unit commitment considering combined electricity and district heating networks," *IEEE Transactions on Sustainable Energy*, vol. 7, no. 2, pp. 480-492, 2015.
- [37] Z. Pan, Q. Guo, and H. Sun, "Feasible region method based integrated heat and electricity dispatch considering building thermal inertia," *Applied Energy*, vol. 192, pp. 395-407, 2017.
- [38] H. Sun *et al.*, "Integrated energy management system: concept, design, and demonstration in China," *IEEE Electrification Magazine*, vol. 6, no. 2, pp. 42-50, 2018.
- [39] Y. Lang, X. Ma, X. Yang, S. Shi, H. Mao, and Y. Chen, "Robust State Estimation of Electricity-Gas Integrated Energy System Based on Multi-Time Scale Data Fusion," in *IEEE 5th Conference on Energy Internet and Energy System Integration (EI2)*, 2021, pp. 1584-1589.
- [40] S. Zheng, J. Liu, Y. Chen, and B. Qi, "Bilinear robust state estimation based on weighted least absolute value for integrated electricity-gas system," *Power System Technology*, vol. 43, no. 10, pp. 3733-3744, 2019.
- [41] Q. Chen, D. Yang, Y. Wang, C. Rehtanz, and H. Pandžić, "Robust state estimation of electricity-gas-heat integrated energy system based on the bilinear transformations," *IET Generation, Transmission & Distribution*, vol. 15, no. 1, pp. 149-163, 2021.
- [42] A. Gomez-Exposito, C. Gomez-Quiles, and A. de la Villa Jaen, "Bilinear power system state estimation," *IEEE Transactions on Power Systems*, vol. 27, no. 1, pp. 493-501, 2011.
- [43] C. Gomez-Quiles, H. A. Gil, A. de la Villa Jaen, and A. Gomez-Exposito, "Equality-constrained bilinear state estimation," *IEEE Transactions on Power Systems*, vol. 28, no. 2, pp. 902-910, 2012.
- [44] J. Zhao, S. Wang, R. Huang, Y. Liu, and Z. Huang, "Robust adaptive decentralized dynamic state estimation with unknown control inputs using field PMU

- measurements," in *IEEE Power & Energy Society General Meeting (PESGM)*, 2020, pp. 1-5.
- [45] J. Zhao *et al.*, "Power system dynamic state estimation: Motivations, definitions, methodologies, and future work," *IEEE Transactions on Power Systems*, vol. 34, no. 4, pp. 3188-3198, 2019.
 - [46] C. Lu, *Numerical methods in fluid mechanics*. Beijing: Science Press, 2014.
 - [47] X. Ma, *Study on Dynamic Characteristics Analysis and Thermal Transient Prediction of Central Heating Pipeline Network*. Taiyuan: Taiyuan University of Technology, 2011.
 - [48] S. Wu and Z. Liu, *Fundamentals and Applications of Computational Fluid Mechanics* Beijing: China Machine Press, 2007.
 - [49] W. Zhang, S. Song, Y. Lin, C. Wang, and Y. Chen, "Multi-source-data state estimation for integrated electricity-heat-building networks," *Sustainable Energy, Grids and Networks*, vol. 36, p. 101131, 2023.
 - [50] S. Song, W. Zhang, Y. Lin, and C. Wang, "Thermo-Electrical state and demand response potential estimation for power systems with building thermostats," *International Journal of Electrical Power & Energy Systems*, vol. 145, p. 108588, 2023.
 - [51] J. Dong, Q. Guo, H. Sun, and Z. Pan, "Research on state estimation for combined heat and power networks," in *IEEE Power and Energy Society General Meeting (PESGM)*, 2016, pp. 1-5.
 - [52] Y. Du, W. Zhang, and T. Zhang, "ADMM-based distributed state estimation for integrated energy system," *CSEE Journal of Power and Energy Systems*, vol. 5, no. 2, pp. 275-283, 2019.
 - [53] Y. Chen, Y. Yao, and Y. Zhang, "A robust state estimation method based on SOCP for integrated electricity-heat system," *IEEE Transactions on Smart Grid*, vol. 12, no. 1, pp. 810-820, 2020.
 - [54] T. Sheng, Q. Guo, H. Sun, Z. Pan, and J. Zhang, "Two-stage state estimation approach for combined heat and electric networks considering the dynamic property of pipelines," *Energy Procedia*, vol. 142, pp. 3014-3019, 2017.
 - [55] S. Ge, X. Liu, L. Ge, H. Liu, and J. Li, "State estimation of regional interconnected Electricity and Gas Networks," *Energy Procedia*, vol. 142, pp. 1920-1932, 2017.
 - [56] H. Zang, M. Geng, M. Huang, Z. Wei, S. Chen, and G. Sun, "Asynchronous and Adaptive State Estimation of Integrated Electricity-Gas Energy Systems," *IEEE Internet of Things Journal*, Early Access, 2022.
 - [57] D. Zhou, S. Ma, D. Huang, H. Zhang, and S. Weng, "An operating state estimation model for integrated energy systems based on distributed solution," *Frontiers in Energy*, vol. 14, pp. 801-816, 2020.
 - [58] L. Chen, X. Hui, Y. Li, M. Huang, and S. Gu, "Dynamic state estimation for integrated natural gas and electric power systems," in *2021 IEEE/IAS Industrial and Commercial Power System Asia (I&CPS Asia)*, pp. 397-402.
 - [59] Y. Chen, Y. Yao, Y. Lin, and X. Yang, "Dynamic state estimation for integrated electricity-gas systems based on Kalman filter," *CSEE Journal of Power and Energy Systems*, vol. 8, no. 1, pp. 293-303, 2020.
 - [60] L. Man, T. Wang, X. Pan, and Y. Li, "State estimation for integrated energy system containing electricity, heat and gas," in *2018 2nd IEEE Conference on Energy Internet and Energy System Integration (EI2)*, pp. 1-6.

- [61] T. Zhang, Z. Li, Q. Wu, and X. Zhou, "Decentralized state estimation of combined heat and power systems using the asynchronous alternating direction method of multipliers," *Applied Energy*, vol. 248, pp. 600-613, 2019.
- [62] G. Yin, D. Četenović, V. Levi, H. Sun, and V. Terzija, "Online State Estimation of the Integrated Electricity and Gas System Based on the Gaseous Circuit Method," *IEEE Transactions on Smart Grid*, Early Access, 2022.
- [63] Y. Lang, X. Ma, X. Yang, S. Shi, H. Mao, and Y. Chen, "Robust State Estimation of Electricity-Gas Integrated Energy System Based on Multi-Time Scale Data Fusion," in *2021 IEEE 5th Conference on Energy Internet and Energy System Integration (EI2)*, pp. 1584-1589.
- [64] Y. Chen, S. Zheng, N. Yang, X. Yang, and K. Liu, "Robust state estimation of electric-gas integrated energy system based on weighted least absolute value," *Autom. Electr. Power Syst*, vol. 43, no. 013, pp. 61-70, 2019.
- [65] H. A. Behrooz and R. B. Boozarjomehry, "Distributed and decentralized state estimation in gas networks as distributed parameter systems," *ISA Transactions*, vol. 58, pp. 552-566, 2015.
- [66] İ. Durgut and M. K. Leblebicioğlu, "State estimation of transient flow in gas pipelines by a Kalman filter-based estimator," *Journal of Natural Gas Science and Engineering*, vol. 35, pp. 189-196, 2016.
- [67] A. Turk, Q. Wu, M. Zhang, and J. Østergaard, "Day-ahead stochastic scheduling of integrated multi-energy system for flexibility synergy and uncertainty balancing," *Energy*, vol. 196, p. 117130, 2020.
- [68] J. Mi and M. E. Khodayar, "Operation of natural gas and electricity networks with line pack," *Journal of Modern Power Systems and Clean Energy*, vol. 7, no. 5, pp. 1056-1070, 2019.
- [69] G. Yin, B. Wang, H. Sun, Q. Guo, and Z. Pan, "Multi-scene adaptive online state estimation of multi-energy network: Development and application," in *Proceedings of the CSEE*, 2020, vol. 40, no. 21, pp. 6794-6804.
- [70] X. Xu, Q. Lyu, M. Qadrdan, and J. Wu, "Quantification of flexibility of a district heating system for the power grid," *IEEE Transactions on Sustainable Energy*, vol. 11, no. 4, pp. 2617-2630, 2020.
- [71] A. Pal, G. A. Sanchez-Ayala, V. A. Centeno, and J. S. Thorp, "A PMU placement scheme ensuring real-time monitoring of critical buses of the network," *IEEE Transactions on Power Delivery*, vol. 29, no. 2, pp. 510-517, 2013.
- [72] A. Pal, A. K. S. Vullikanti, and S. S. Ravi, "A PMU placement scheme considering realistic costs and modern trends in relaying," *IEEE Transactions on Power Systems*, vol. 32, no. 1, pp. 552-561, 2016.
- [73] A. Srivastava and S. Maheswarapu, "Optimal PMU placement for complete power system observability using binary cat swarm optimization," in *2015 International Conference on Energy Economics and Environment (ICEEE)*, 2015: IEEE, pp. 1-6.
- [74] S. Chakrabarti and E. Kyriakides, "Optimal placement of phasor measurement units for power system observability," *IEEE Transactions on Power Systems*, vol. 23, no. 3, pp. 1433-1440, 2008.
- [75] B. Azimian, R. S. Biswas, S. Moshtagh, A. Pal, L. Tong, and G. Dasarathy, "State and topology estimation for unobservable distribution systems using deep

- neural networks," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1-14, 2022.
- [76] Y. Chen, Y. Gao, J. Zhao, and L. Ge, "Review on integrated energy system state estimation," *High Voltage Engineering*, vol. 47, no. 7, pp. 2281-2292, 2021.
- [77] Y. Xiaonan, L. Yansheng, and Y. Yuan, "Observability analysis on state estimation of integrated electricity-gas system," *Electric Power*, vol. 53, no. 10, pp. 133-139, 2020.
- [78] J. Zhao *et al.*, "Roles of dynamic state estimation in power system modeling, monitoring and operation," *IEEE Transactions on Power Systems*, vol. 36, no. 3, pp. 2462-2472, 2020.
- [79] F. F. Wu and A. Monticelli, "Network observability: theory," *IEEE Transactions on Power Apparatus and Systems*, no. 5, pp. 1042-1048, 1985.
- [80] A. Monticelli and F. F. Wu, "Network observability: Identification of observable islands and measurement placement," *IEEE Transactions on Power Apparatus and Systems*, no. 5, pp. 1035-1041, 1985.
- [81] B. Gou, "Jacobian matrix-based observability analysis for state estimation," *IEEE Transactions on Power Systems*, vol. 21, no. 1, pp. 348-356, 2006.
- [82] G. Krumpholz, K. Clements, and P. Davis, "Power system observability: a practical algorithm using network topology," *IEEE Transactions on Power Apparatus and Systems*, no. 4, pp. 1534-1542, 1980.
- [83] R. R. Nucera and M. L. Gilles, "Observability analysis: a new topological algorithm," *IEEE Transactions on Power Systems*, vol. 6, no. 2, pp. 466-475, 1991.
- [84] A. Monticelli, A. Garcia, and I. Slutsker, "Handling discardable measurements in power system state estimation," *IEEE Transactions on Power Systems*, vol. 7, no. 3, pp. 1333-1340, 1992.
- [85] A. Rouhani and A. Abur, "Observability analysis for dynamic state estimation of synchronous machines," *IEEE Transactions on Power Systems*, vol. 32, no. 4, pp. 3168-3175, 2016.
- [86] G. Wang, C. C. Liu, N. Bhatt, E. Farantatos, and M. Patel, "Observability of nonlinear power system dynamics using synchrophasor data," *International Transactions on Electrical Energy Systems*, vol. 26, no. 5, pp. 952-967, 2016.
- [87] J. Qi, K. Sun, and W. Kang, "Adaptive optimal PMU placement based on empirical observability gramian," *IFAC-PapersOnLine*, vol. 49, no. 18, pp. 482-487, 2016.
- [88] J. Qi, K. Sun, and W. Kang, "Optimal PMU placement for power system dynamic state estimation by using empirical observability Gramian," *IEEE Transactions on Power Systems*, vol. 30, no. 4, pp. 2041-2054, 2014.
- [89] Z. Li, W. Zheng, J. Zheng, J. Zhao, and Q. Wu, "Observability Analysis of Integrated Electricity and Heating Systems with Thermal Quasi-dynamics in Pipelines," *Accepted in CSEE Journal of Power and Energy Systems*, 2022.
- [90] O. Romay, J. L. Sánchez-Garduño, and C. R. Fuerte-Esquivel, "Unified state estimation in integrated natural gas and electricity networks," in *2020 IEEE International Autumn Meeting on Power, Electronics and Computing (ROPEC)*, 2020, vol. 4: IEEE, pp. 1-6.

- [91] K. Yang, X. Shen, Q. Guo, and H. Sun, "A Numerical Observability Analysis Method for Combined Electric-Gas Networks," in *2018 2nd IEEE Conference on Energy Internet and Energy System Integration (EI2)*, 2018: IEEE, pp. 1-9.
- [92] J. Dong, H. Sun, Q. Guo, T. Sheng, and Z. Qiao, "State estimation of combined electric-gas networks for energy internet," *Power System Technology*, vol. 42, no. 2, pp. 400-408, 2018.
- [93] J. Dong, H. Sun, Q. Guo, and Z. Pan, "State estimation for combined electricity and heat networks," *Power System Technology*, vol. 40, no. 6, pp. 1635-1641, 2016.
- [94] X. Yang, B. Sun, F. Han, and Y. Lang, "A bilinear robust state estimation method for electric thermal integrated energy system," in *Journal of Physics: Conference Series*, 2020, vol. 1549, no. 5: IOP Publishing, p. 052085.
- [95] Y. Chen, S. Zheng, N. Yang, and Y. Lang, "Bilinear robust state estimation method for integrated electricity-heat energy systems," *Electr. Power Autom. Equip*, vol. 39, no. 8, pp. 47-54, 2019.
- [96] X. Yang, Y. Lang, H. Zhang, Y. Lv, C. Zhu, and Y. Liu, "A Two-layer Sequential State Estimation Method for electricity and heat Networks," in *Journal of Physics: Conference Series*, 2021, vol. 1802, no. 3: IOP Publishing, p. 032088.
- [97] T. Zhang, Z. Li, Q. Wu, and X. Zhou, "Dynamic state estimation of combined heat and power system considering quasi-dynamics of temperature in pipelines," in *2018 International Conference on Power System Technology (POWERCON)*, 2018: IEEE, pp. 232-237.
- [98] D. Lei, W. Chunfei, and L. Ye, "Multi-snapshot state estimation of electric-gas integrated energy system," *Power System Technology*, vol. 44, no. 9, pp. 3458-3465, 2020.
- [99] T. Zhang, W. Zhang, Q. Zhao, Y. Du, J. Chen, and J. Zhao, "Distributed real-time state estimation for combined heat and power systems," *Journal of Modern Power Systems and Clean Energy*, vol. 9, no. 2, pp. 316-327, 2020.
- [100] X. Liu, Y. Li, Q. Sun, and Y. Pan, "Interaction and joint state estimation of electric-gas-thermal coupling network," *Power Syst. Technol*, vol. 49, no. 2, pp. 479-490, 2021.
- [101] S. Chen, Z. Wei, G. Sun, H. Zang, Y. Zhang, and Z. Ma, "An interval state estimation for electricity-gas urban energy systems," in *2017 IEEE Conference on Energy Internet and Energy System Integration (EI2)*, 2017: IEEE, pp. 1-5.
- [102] C. Wang, M. Geng, Q. Xu, and H. Zang, "A Fast State Estimator for Integrated Electrical and Heating Networks," *Energies*, vol. 13, no. 17, p. 4488, 2020.
- [103] W. Zheng, Z. Li, X. Liang, J. Zheng, Q. Wu, and F. Hu, "Decentralized state estimation of combined heat and power system considering communication packet loss," *Journal of Modern Power Systems and Clean Energy*, vol. 8, no. 4, pp. 646-656, 2020.
- [104] Y. Lin and A. Abur, "A new framework for detection and identification of network parameter errors," *IEEE Transactions on Smart Grid*, vol. 9, no. 3, pp. 1698-1706, 2016.
- [105] Y. Lin and A. Abur, "Robust state estimation against measurement and network parameter errors," *IEEE Transactions on Power Systems*, vol. 33, no. 5, pp. 4751-4759, 2018.

- [106] Y. Lin and A. Abur, "Fast correction of network parameter errors," *IEEE Transactions on Power Systems*, vol. 33, no. 1, pp. 1095-1096, 2017.
- [107] PJM Manual 01, "Control Center and Data Exchange Requirements," System Operations Divisions, 2014.
- [108] Technical Report PES-TR105, "Towards Integrating Cyber and Physical Security for a More Reliable, Resilient, and Secure Energy Sector," IEEE/NERC Joint Task Force on Security Integration into BPS Engineering Practices, December 2022.
- [109] H. Sun, Q. Guo, W. Wu, B. Wang, T. Xia, and B. Zhang, "Integrated energy management system with multi-energy flow for Energy Internet: Design and application," *Automation of Electric Power Systems*, vol. 43, no. 12, pp. 122-128, 2019.
- [110] Z. Pan, Q. Guo, and H. Sun, "Interactions of district electricity and heating systems considering time-scale characteristics based on quasi-steady multi-energy flow," *Applied Energy*, vol. 167, pp. 230-243, 2016.